

Universal Surprise-Minimizing Agents

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Abstract

We formally analyze a computationally unbounded agent whose goal is surprise minimization on incoming observations, within the framework of Universal AI. This objective is described by the free energy principle, which we reformulate for this setting. Having removed the problem of computational constraints, we investigate whether the sole goal of surprise minimization leads to interesting or trivial behavior. As we will see, this heavily depends on the (Kolmogorov) complexity of the environment. We show that the optimal behavior for the agent is to enter a niche of its world that can be easily modeled, without altering the prior weight on simple environments significantly. Nonetheless, we prove bounds for the complexity of this niche and how it relates to the dimensionality of the action and observation spaces, horizon, and complexity of the true environment. This bound entails a counter-intuitive short term exploratory behavior and information gain in pursuit of reaching the niche. Additionally, we argue how the agnostic agent, being intrinsically suboptimal, might explore beyond the optimal niche. We demonstrate both asymptotic optimality and sub-linear regret of our agent. Finally, we motivate how it controls the probability of dying, as opposed to the Knowledge-Seeking Agent (KSA), which, in contrast, maximizes surprise.

1 Introduction

We seek to explore how a universal agent, analogous to AIXI¹ [Hut24], behaves when it is given the intrinsic goal of surprise minimization on incoming observations. The latter is described by the Free Energy Principle (FEP) [Fri23] and active inference [Da 20]. Nonetheless, it is implemented in only a few studies in RL [Ber21; Zha22].

On the other hand, within Universal AI (UAI), an opposite kind of intrinsic motivation, of entropy seeking, has been proposed with the KSA [Ors13; Ors14].

Trying to reconcile these two objectives, our main hypothesis is that exploratory behavior would naturally emerge from surprise minimization in certain environments. Furthermore, we hypothesize that exploration would be far more moderate than the one performed by the KSA, allowing for avoidance of death.

The strength of the UAI theoretical framework, which is not tied to computational constraints, lies in the fact that few parameters fully define the model. In particular, on the agent’s side, we only need the prior beliefs, the reward function, the horizon function and the action and observation spaces. We deem it interesting to explain how these parameters, including the true environment, interact with each other, rather than proposing surprise minimization as an ideal objective to be applied to *any* agent. In fact, it can be an undesired goal in

certain environments, as it can lead, as we will see, to minimal exploration (Sec. 6 and 7).

With this, we turn the problem of exploration and sub-optimality of AIXI [Ors10; Hut24] on its head by embracing the potential “traps” that in a regular setting would be considered sub-optimal and thus preferably avoided. Here, instead, *safe* niches are exactly what the agent seeks. As we will demonstrate, the resulting behavior is not necessarily trivial and heavily depends on the (Kolmogorov) compressibility of the observations produced by the environment (Sec. 8). In fact, the agent has to exhibit an exploratory behavior to find a niche first, especially in those *complex* environments whose dynamics evolve through time and a static niche is harder to find².

This is in line with the theories on instrumental convergence: given an arbitrary goal, an agent has to behave intelligently to obtain instrumental resources and ensure basic survival, in order to achieve it. This phenomenon emerges in the AIXI setting too as shown in [Rin11], where different goals lead to similar “survival” agents that prioritize their survival when death, treated as a modification of their optimal program, is possible.³

With this, in a sense, we strip down the goal of the agent to its own survival, which is already implied by different reward functions, here encoded by surprise minimization⁴. The latter

¹In short, AIXI is defined as an ideal theoretical agent which maximizes reward considering all possible futures predicted by a Bayesian mixture over a countable class of environments, $\mathcal{M}$. This is usually the class of (lower semi-)computable environments, which is more extensive than the class of Markov environments often used in RL. For each new observation, the exact posterior update is performed. The initial beliefs are set by Solomonoff prior, which exponentially favors environments with lower Kolmogorov complexity.

²Think of a real world application, such as a job finder agent. It might be hard for it to find a niche of the job market which changes little with time.

³Their program is optimal because it modifies their surrounding to have a “delusion box” which maximizes rewards with artificial observations.

⁴We could use a simpler reward, such as assigning +1 for each time step during which the agent has survived. However, this might render the agent short-sighted when combined with a horizon shorter than its potential lifetime. On the other hand, surprise minimization, while being supported by accounts on real biological life (FEP), can underwrite

can also be expressed as minimization of prediction error, or as maximization of the likelihood of new observations. As detailed in Sec.9.4, the latter has a connection with AIXI’s semi-measure death too.

Finally, we demonstrate asymptotic optimality (Sec. 9.2) and sublinear regret (Sec. 9.3) in finite complexity environments. These are usually two mutually exclusive properties in AIXI, as they concern long-term and short-term optimality, respectively. Altogether, this makes the universal Surprise-Minimizing Agent (SMA) an interesting case.

2 Notation

The reader may skim this section and come back to it whenever a new or unfamiliar symbol is encountered.

2.1 FEP Framework

To discuss more generally about our agent and its embeddedness from the perspective of the free energy principle, we use the following notation.

True distribution (p) We define the dynamics of the trajectories of our system with the joint distribution p , describing the dynamics of four particles: external state η , observation o , action a and the internal state (of the agent) w .

$$p(\eta_{1:t}, o_{1:t}, a_{1:t}, w_{1:t}) \quad (1)$$

As a shorthand, we often omit the subscript $1:t$ indicating the beginning and end time step of a discrete trajectory: $x = x_{1:t} \in \mathcal{X}^t$.

Agent’s model (q) Conversely, the agent’s beliefs about these dynamics are represented by a distribution q over the same variables. Internal states w parametrize the posterior beliefs over external states: $q(\eta \mid o, a, w) = q_w(\eta \mid o, a)$.

2.2 Interaction Histories

Interaction pair (h_t)

Action and observation pair at time t :

$$h_t := a_t o_t \in \mathcal{H} := \mathcal{A} \times \mathcal{O}.$$

Where $a_t \in \mathcal{A}$ is action taken by the agent at time t , and $o_t \in \mathcal{O}$ is the observation generated by the environment in response.

History segment ($h_{i:s}$)

Sequence of interaction pairs representing a segment of history from indices i to s :

$$h_{i:s} \in \mathcal{H}^{i-s+1}.$$

History ($h_{<t} = h_{1:t-1}$)

History of previous interaction cycles up to the *current* time step t .

$$h_{<t} = \langle a_1, o_1, a_2, o_2 \dots a_{t-1}, o_{t-1} \rangle$$

Often simply denoted as h . To append an action a_t to history $h_{<t}$ we write $h_{<t} a_t$.

Action and observation sequences ($a_{<t}, o_{<t}$)

long-term survival. In fact, as a byproduct of moving to simpler (less unpredictable) areas of the environment, the agent empowers itself to do so again in future time-steps, when new unforeseen circumstances can arise.

They represent the actions and observations sequences, respectively, within a given history $h_{<t}$; e.g.:

$$a_{<t} = \langle a_1, a_2, \dots, a_{t-1} \rangle$$

Often the subscript is left implicit: $a_{<t} = a$ and $o_{<t} = o$.

2.3 Discounting and Horizon

Discount (γ_t)

Discount at time t .

Horizon (m)

Horizon $m \in \mathbb{N}^+$, represents the number of future time-steps the agent considers.

2.4 Environment Class

Chronological Turing machines ($T, U, \dots$) Chronological TMs defined as in [Hut24, Def. 7.4.4]. Given a program p , they take an action stream $a_{1:m}$ as input on a separate tape and output observations $o_{1:m}$. Each observation o_k must be printed prior to reading a_{k+1} :

$$T(p, a_{1:m}) \rightarrow o_{1:m} \text{ is true iff } T(p, a_{1:k}) = o_{1:k} \forall k \leq m$$

Kolmogorov complexity (K)

The prefix Kolmogorov complexity of x is defined as the length of the minimal program $\in \mathbb{B}^*$ which outputs x when executed on a reference prefix (chronological) Universal Turing Machine (UTM) U . We can optionally provide U with side information y or action stream a . Hence, we provide the general definition of conditional prefix complexity below.

$$K(x \mid y, a) = \min_{p \in \mathbb{B}^*} \{l(p) : U(y'p, a)\}$$

Both y and a can be empty strings.

Model class ($\mathcal{M}$) and environmental distribution ($\nu, \rho, \varrho, \dots$)

Countable class of environments $\nu \in \mathcal{M}$. In the following we generally consider $\mathcal{M}_{det}$ as the class of all lower semi-computable deterministic semi-measures. Each model ν_p within this class can be enumerated as a program $p \in \mathbb{B}^*$, which is run on a chronological [Hut24, Def. 7.4.4] UTM U and prints percept o_t on the output tape, given actions $a_{\leq t}$ in the “action tape”:

$$\nu_p(o_t \mid h_{<t} a_t) = \begin{cases} 1 & \text{if: } U(p, a_{\leq t}) \rightarrow o_{\leq t} \\ 0 & \text{otw} \end{cases}$$

We might have $\sum_{o_t} \nu_p(o_t \mid h_{<t} a_t) = 0 < 1$ if p never halts, making ν_p a semi-measure, since the probabilities do not sum to one.

True environment (μ)

True environment, $\mu \in \mathcal{M}_{det}$.

2.5 Chronological Measures

Chronological distribution of histories ($P_\rho^\pi(h)$)

Chronological measure over interaction history $h \in \mathcal{H}^t$, with environmental predictor ρ and policy π :

$$P_\rho^\pi(h) := \prod_{k=1}^t \pi(a_k \mid h_{<k}) \rho(o_k \mid h_{<k} a_k)$$

with $h_{<1} = \epsilon$, the empty history.

Chronological distribution of observations ($\varrho(o_{1:t} \parallel a_{1:t})$)

Chronological distribution on observations $o_{1:t}$ conditioned on the action sequence $a_{1:t}$, given distribution ϱ on next observation o_i :

$$\varrho(o_{1:t} \parallel a_{1:t}) = \prod_{i=1}^t \varrho(o_i \mid h_{<i} a_i).$$

2.6 Universal Mixture

Solomonoff prior weight (w_ν)

Solomonoff prior weight on environment ν with Kolmogorov complexity $K(\nu)$:

$$w_\nu = 2^{-K(\nu)}.$$

Posterior weight ($w_\nu(h_{<t})$)

The posterior weight of environment ν , given an initial prior w_ν :

$$w_\nu(h_{<t}) = \frac{w_\nu \nu(o_{<t} \parallel a_{<t})}{\xi(o_{<t} \parallel a_{<t})} = \frac{w_\nu P_\nu^\pi(h)}{P_\xi^\pi(h)}.$$

Universal mixture (ξ)

ξ is the Bayesian mixture of all the environmental models within $\mathcal{M}$ (Sec. 2.4), with prior w_ν defined above. Therefore, the likelihood of observation o_t at time t is defined as follows.

$$\xi(o_t \mid h_{<t} a_t) = \sum_{\nu \in \mathcal{M}_{det}} 2^{-K(\nu)} \nu(o_t \mid h_{<t} a_t) \quad (2)$$

This mixture is similar to the agentic version of Solomonoff prior [Hut24, Thm. 7.4.5]. ξ is lower semi-computable. In fact, $K(\nu_p) = K(p)$ [Hut24, Def. 2.7.19] is upper semi-computable and therefore $2^{-K(p)}$ is lower semi-computable; while each ν_p is lower semi-computable too.

2.7 Policies

Policy (π)

A policy which selects the next action, given an interaction history of length m or shorter:

$$\pi : \mathcal{H}^{\leq m} \rightarrow \Delta(\mathcal{A}), \quad \mathcal{H}^{\leq m} := \bigcup_{t=0}^m \mathcal{H}^t.$$

2.8 Others

Additive inequalities ($\stackrel{\pm}{\leq}$) From [Hut24, Def. 2.7.3] we borrow the following notation for (in)equalities within additive constants, which do not depend on x .

$$\begin{aligned} f(x) \stackrel{\pm}{\leq} g(x) &\iff f(x) = g(x) + O(1) \\ f(x) \stackrel{\pm}{=} g(x) &\iff f(x) \stackrel{\pm}{\leq} g(x) \text{ and } f(x) \stackrel{\pm}{\geq} g(x) \end{aligned}$$

Length of a binary string ($l(x)$) Given any object x we refer to $l(x)$ as the length in bits of x . If the latter is not a binary string, we consider the length of its binary representation.

Prefix code ($\mathbf{x}'$) When applied on a binary string $x \in \mathbb{B}^*$, the prime notation indicates a second-order prefix code [Hut24, Eq. 2.1.4] as:

$$\mathbf{x}' := 1^{\ell(\ell(x))} 0^{\ell(x)} x$$

b^n with $b \in \mathbb{B}$ indicates a string where b is repeated n times; e.g. $1^5 = 11111$.

3 Background

There have already been plenty of discussions in the RL literature regarding the Intrinsic Motivations (IM) an agent can have [Ber21; Zha22; Kie25]. These, in fact, are meant to circumvent the problems related to externally assigned rewards. More specifically, they have been usually defined through information theoretic measures [Kie25]. [Biel18] shows the overlap between the active inference framework, different information theoretic IM and universal RL, inspiring the connection we draw here (Sec. 4.2). These intrinsic motivations depend on the internal model of the agent rather than on external contingencies. This constraint effectively removes the “reward function” parameter as rewards now fully depend on the agent’s model, and can adapt to different environments and embodiments.

3.1 Knowledge-Seeking Agents

Within UAI, Orseau et al. analyzed Knowledge Seeking Agents (KSA) which maximize surprise in deterministic environments (Shannon-KSA) [Ors14] and Information Gain (IG) in general stochastic environments (KL-KSA) [Ors13]:

$$r_t^{KSA} = \text{IG}_{h_{<t}}(h_t) := \sum_{\nu \in \mathcal{M}} w_\nu(h_{<t} h_t) \log \left[\frac{w_\nu(h_{<t} h_t)}{w_\nu(h_{<t})} \right] \quad (3)$$

The two are equivalent when the environment is deterministic. On the other hand, surprise maximization has trivial behavior in the stochastic setting as the agent can get stuck in the noisy parts of the environment. Indeed, since the produced observations cannot be predicted, they provide maximal surprise.

The strength of KL-KSA, which we will refer to simply as KSA, is that it is demonstrated to gather complete (off-policy) knowledge of the true environment μ [Ors13, Th. 7].

3.2 Extrinsic Rewards and Death

The reward specified in Eq. 3 gives an asymptotically optimal agent even when combined with an extrinsic reward, as shown with InQ [Coh19]; that is, an agent that alternates between the regular AIXI policy, maximizing the extrinsic reward, and the KSA (exploratory) policy, with a decreasing probability of using the latter. However, in general, maximizing information gain, while guaranteeing optimality, leads to almost-sure death or incapacitation in those environments where such outcomes are possible. Indeed, KSA-based agents have been shown to try every finite sequence of actions at least once [Coh21], including those that would kill the agent. An additional downside is that extrinsic rewards still have to be externally defined and traditional exploitation can be implemented only through hybrid agents with two separate goals such as InQ or BayesExp [Hut24, Sec. 9.4]. On the other hand, with a surprise-minimizing agent, extrinsic preferences may be incorporated in the prior on observations, without affecting the simplicity and elegance of the AIXI algorithm (see the discussion in Sec. 10).

3.3 Scientific Behavior and Curiosity

One of the main motivations behind KSA was to implement an agent that could explain or exhibit optimal scientific behavior. This is interpreted by Orseau as maximizing the information that the agent gets out of its environment. This information

is encoded by the movement of beliefs regarding the true environment (the posterior). Since the ξ predictor computes the posterior in a Bayesian and optimal fashion, the agent cannot “reward-hack” its model to induce information gain which does not lead to an improved comprehension of its surroundings. This means that seeking IG pushes, in the limit, all the posterior weight on the true environment or equivalent models. Yet, bounded approximations of KSA might have critical issues while searching for information gain. In fact, with limited model capacity and precision, they would not be able to compute the exact Bayesian weight for a vast class of environments. In fact, if the agent can only consider a limited hypothesis class, the same models could be cyclically discarded and re-added to the model pool, allowing for an endless stream of IG.

Moreover, KSA does not take actions to facilitate the job of understanding its surroundings; for example, it does not move towards parts of the environment easier to model, which would be crucial for bounded agents. This is, instead, achieved by surprise-minimizing agents, which ideally leave “safe areas” only when this is helpful to improve their internal model, if this is necessary at all. Such agents should seek to improve the (future) predictive model of their local surrounding, similarly to the algorithm proposed by Schmidhuber [Sch06] to model a curiosity driven agent. The latter is rewarded by the decrease in prediction error over past observations, on which the agent is re-iteratively scored at each time step. Therefore, this agent doesn’t necessarily look for *any* kind of novel information, but focuses on the insights it could organically build upon given the current model. Nonetheless, this performance measure might induce overfitting of the past, since the prediction error on the whole history is recomputed at each time step.⁵

On the other hand, the agent proposed here uses a loss function which depends only on future or current observations.⁶

	Maximization	Minimization
IG	KL-KSA	KMA
Surprise	Shannon-KSA	SMA

Table 1: Scheme of the cited agents and their objectives. SMA corresponds to the surprise-minimizing agent presented here.

Finally, Table 1 summarizes the relationship between the objective of the surprise-minimizing agent (SMA) proposed here and KSA. The matrix reveals an additional Knowledge Minimizing Agent (KMA), as we will show, this agent is equivalent to SMA in the deterministic setting. In fact, while expected IG is related to the KL divergence between the true likelihood of observations and the agent’s likelihood, expected surprise is a cross-entropy between the two; the two quantities coincide when the entropy of the true distribution vanishes, which is precisely the deterministic case (Sec. 6).

⁵Moreover, AIXI agents already implicitly enforce a perfect fit of the past in deterministic setting: only the environments consistent with the history are kept in posterior. Another key element of Schmidhuber’s construction is that it controls (minimizes) for the complexity of the model, which, as we will see, is implicitly done by the surprise-minimizing agent portrayed here.

⁶We don’t use the difference in prediction error, but directly the latter as they are functionally equivalent.

4 The Free Energy Principle

In this section we reformulate the free energy principle to fit with the discrete, sequential setting of UAI and its assumptions. This ultimately serves to specify the right cost function (or negative reward) for our agent (Sec. 5). Firstly, it is crucial to note that the free energy principle is usually formulated on hidden Markov processes, where the latent state evolves with time. In UAI, by contrast, we model a static (hidden) environment μ which does not vary, since it describes the entire world and can be fully defined by a Turing machine. This entails that, while in FEP the uncertainty is on both the hidden state and the dynamics of the environment, the uncertainty here is only on one of the two, as state and dynamics are condensed into a single variable. We will alternatively refer to this variable as η or $\nu \in \mathcal{M}$ when we assume the latter belongs to a specific class of environments $\mathcal{M}$. To summarize:

Assumption 4.1 (Hidden causes). The hidden cause/state η fully describes the environment dynamics and therefore doesn’t change with time $\forall t : p(\eta_t) = p(\eta_{t+1})$.

Assumption 4.2 (True environment). Under the true distribution p , only the true environment μ has all the prior probability mass:

$$p(\eta_0) = \begin{cases} 1 & \text{if } \eta_0 = \mu \\ 0 & \text{otw} \end{cases} \quad (4)$$

4.1 Surprise Minimization

The objective of surprise minimization stems from a first principle account of embedded agents. These can be modeled as a particle in multi-dimensional space, whose dynamics can be defined by a set of stochastic differential equations. In this setting we have four particles interacting with each other (Fig. 1): internal states w , active states a , sensory states o and external states η , representing the causes hidden to the agent. Their true dynamics are represented by distribution p (Sec. 2.1). By definition, we have that the trajectories which maximize p are the ones that are most likely to occur, and will *surely* occur in a deterministic system⁷ As a matter of fact, we assume the following.

Assumption 4.3. The dynamics of our system, $p(\eta, o, a, w)$, are deterministic.

From this it follows:

$$(\eta, o, a, w) = \arg \max p(\eta, o, a, w) \quad (5)$$

$$\stackrel{(a)}{=} \arg \min -\log [p(\eta, o, a, w)] \quad (6)$$

(a) holds because of monotonicity of the logarithm.⁸

Another key assumption, which is at the basis of any FEP derivation, is:

⁷For a classical derivation in continuous and stochastic dynamics we refer to [Fri231] and [Fri232].

⁸We apply the logarithm throughout when minimizing/maximizing the value of a probability distribution for its additive properties, which allow us to isolate the contribution of each time-step. Moreover, they allow us to express formulas in terms of information theoretic quantities, which in turn behave nicely with algorithmic theory and Kolmogorov complexity (e.g. Thm. 7.6). Finally, it has also been historically used in free energy literature as it is necessary for the path integral formulation of dynamical systems, and it allows to express things in term of energy-related quantities.

Assumption 4.4 (Markov blanket). External states can only interact with internal states through sensory states, and internal states can influence the world only through active states.

We consider this a relatively weak assumption which underlies most accounts of agency. Assumption 4.4 entails a conditional independence between the agent and the environment:

$$(a, w) \perp \eta \mid o \iff p(\eta, o, a, w) = p(o, a, w) \quad (7)$$

It follows that the agent should minimize surprise with respect to the true generative model $p(o, a, w)$, where the external states have been marginalized out. Indeed, combining Eq. 7 with Eq. 6 we have

$$(a, w) = \arg \min -\log p(o, a, w) \quad (8)$$

Since, from the perspective of the agent, we do not know the true posterior $p(\eta \mid o, a, w) = \frac{p(\eta, o, a, w)}{p(o, a, w)}$, we cannot compute $p(o, a, w) = \frac{p(\eta, o, a, w)}{p(\eta \mid o, a, w)}$. We will therefore adopt a variational approach.

Importantly, we have theoretical Bayesian posterior calculation capabilities, contrary to practical implementations of variational inference. However, we cannot compute the *true* posterior, nor the joint distribution, because we do not know the true prior on external causes $p(\eta)$ (Eq. 4), which represents the state of affairs of the external world at the beginning of the agent’s life. This is a natural assumption for an agent which needs to eventually learn about it. Still, since the hidden cause η fully describes the environment, meaning it describes how observations are generated too, the generative model of the agent conditioned on a specific η matches the true one.

Assumption 4.5 (Generative model). We assume that the agent’s model q includes the true generative model conditioned on η .

$$q(o \mid a, w, \eta) = p(o \mid a, w, \eta) \quad (9)$$

However, the priors may differ: $q(\eta) \neq p(\eta)$.

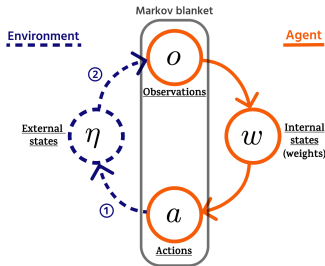


Figure 1: Minimal formulation of SMA’s causal model⁹. It reflects the cybernetic model of universal RL (As. 5.3), with two intermediary variables in the interaction cycle between observations and actions: external states η and internal states w . These can be thought as two Turing machines (storing the whole history on input and work tapes) representing the environment and the agent’s policy, respectively (as in [Hut07]). Alternatively, we can describe the internal states as the weights of the ξ model. In this case, we would have to add an additional relationship, $o \rightarrow a$, to capture the full expressivity of the cybernetic model, which allows for policies that do not depend solely on the weights, but on the full history.¹⁰

Looking at Fig. 1, As. 4.5 means that the causal relationship ② is known. Since actions and internal states depend only on observations, and the solid line relationships between them (arrows in orange) are assumed to be known (because they are internal to the agent), we can obtain $q(o, a, w, \mid \eta) = p(o, a, w, \mid \eta)$. Nonetheless, we need $p(\eta)$ in order to compute $p(o, a, w) = \sum_{\eta} q(o, a, w, \mid \eta)p(\eta)$.

As. 4.5 relies on As. 4.1 and on the assumption that, coherently with UAI’s framework, the agent has no computational constraints, and thus can store and run the generative model of any possible environment, fully described by η .

As already mentioned, this represents the main clash with classical FEP literature, where the hidden causes have multiple degrees of freedom and they evolve with time.

In the discussion section we will further motivate our approach, and point to potential ways to represent the environment in a coarser and local manner.

$$q(\eta_0 = \nu) = 2^{-K_{\nu}} = w_{\nu} \quad (10)$$

4.2 Surprise Minimization From The Perspective of The Agent

Since the posteriors of p and q differ, the agent can minimize *expected* surprise by minimizing the variational bound on $-\log p(o, a, w)$, based on the FEP.

Theorem 4.1 (Expected Surprise Minimization). *Given assumptions 4.1, 4.4 and 4.5, minimizing the variational bound on expected surprise over future trajectories o, a, w , governed by true dynamics $p(o, a, w, \eta)$, is equivalent to the minimization of expected surprise with respect to $q(o, a, w)$. Given this objective, the universal mixture ξ is balanced Pareto optimal with respect to Solomonoff’s prior.*

$$\arg \min_{a, w} \mathbb{E}_{o \sim p(\cdot \mid a, w)} [F(o, a, w)] = \min_{\pi} \mathbb{E}_{P_{\pi}^{\pi}(h)} [-\log P_q^{\pi}(h)] \quad (i)$$

$$\begin{aligned} & \min_{q, \pi} \mathbb{E}_{\mu \sim \xi(\cdot)} \mathbb{E}_{h \sim P_{\mu}^{\pi}(h)} [-\log P_q^{\pi}(h)] \\ & = \min_{\pi} \mathbb{E}_{\mu \sim \xi(\cdot)} \mathbb{E}_{h \sim P_{\mu}^{\pi}(h)} [-\log P_{\xi}^{\pi}(h)] \end{aligned} \quad (ii)$$

Proof. The proofs are given in Sec. A of the appendix. $\square$

⁹Cycles are allowed since relationships are temporal, meaning that each arrow $x \rightarrow y$ establishes sequential relationships $x_t \rightarrow y_{t+n}$ with $n \in \mathbb{N}$.

¹⁰We might want to also add an arrow $a \rightarrow o$ if η represents only the program of the environment without the current state of the input/work tapes. Altogether these additional two arrows $o \rightleftharpoons a$ would give us the same causal model displayed in [Fri232, Fig. 2].

5 Surprise Minimization Within UAI

As anticipated, we define our agent within the universal history-based RL framework. This setting relies on the following additional assumptions.

Assumption 5.1 (Computable environment). The true environment μ belongs to the model class $\mathcal{M}$ of (lower semi-)computable environments (Sec. 2.4).

Assumption 5.2 (Unbounded computational resources). The agent can consider any possible computable environment, weighting them with an exact Bayesian posterior at every time step. The weights of each environment ν are initially set by Solomonoff prior: $\xi(\nu) = 2^{-K(\nu)}$, which favors "simpler" explanations in accordance with Occam's razor. The agent can also calculate the expected cost precisely by exhaustively considering all possible future interactions.

The interaction model between agent and environment (Fig. 1) is formalized using the standard UAI notation described in Sec. 2.2.

Assumption 5.3 (Cybernetic model). The agent interacts with the environment in discrete cycles according to [Hut24, Def. 6.1.4].

We start from $t = 1$ and empty history $h_{<1}$.

At time t , the agent first selects an action $a_t \sim \pi(\cdot | h_{< t})$, after which the true environment μ responds with an observation $o_t \sim \mu(\cdot | h_{< t} a_t)$. The process repeats iteratively with cycle $t + 1$ after updating the history and policy.

We assume that the agent always has access to the history up to the current time step ($h_{< t}$), and that both the action and observation spaces are finite and discrete.

5.1 The Cost Function

The intrinsic motivation we derived with Thm. 4.1 over interaction trajectories can be expressed per time step, given two generic environmental distributions (Sec. 2.4) ρ and ϱ .

Definition 5.1 (Point-wise Surprise). The per-step surprise is expressed as the negative log-likelihood given by the agent's predictive model ϱ , over the new incoming observation.

$$S_{\varrho}(o_t | h_{< t} a_t) = -\log \varrho(o_t | h_{< t} a_t) \quad (S[o])$$

Alternatively, coherently with Thm. 4.1, we can consider the surprise over the whole interaction pair.

$$\begin{aligned} S_{\varrho}^{\pi}(h_t | h_{< t}) &= -\log P_{\varrho}^{\pi}(h_t | h_{< t}) \\ &= -\log[\pi(a_t | h_{< t}) \varrho(o_t | h_{< t} a_t)] \quad (S[h]) \end{aligned}$$

This latter formulation integrates uncertainty on actions as part of the surprise objective. These two formulations are equivalent given deterministic policies. Moreover, the set of optimal policies always includes a deterministic policy as the one which takes the action with minimum Q-cost (Q-value if we were to maximize reward) at each time step¹¹.

This cost could be mapped to the classical reward r_t , intended to be maximized, by taking the negative: $r_t = -S_{\varrho}(o_t)$. In fact, the only difference from a regular AIXI agent is that here we minimize a cost whose value can be in the interval $[0, \infty)$ instead of maximizing a reward in the interval $[0, 1]$.¹²

¹¹Q-cost can be defined in the same way as Q-value [Hut24, Def. 6.7.1]. Furthermore, [Hut24, Thm. 6.7.2] shows that a (value) function analogous to our expected cost (Eq.12) can be rewritten with Bellman equations if ρ is a proper measure

¹²If we want to maintain the canonical AIXI reward range $r_t \in [0, 1]$,

5.2 The Expected Cost

Definition 5.2. The expected cost (or negative expected value) at time t is the expected discounted¹³ surprise:

$$S_{\rho||\varrho}^{\pi, \gamma}(h_{< t}) := \underbrace{\frac{1}{\Gamma_t}}_{\text{optional normalization}} \mathbb{E}_{\rho}^{\pi} \left[\sum_{k=t}^{\infty} \gamma_k [S_{\varrho}() \text{ or } S_{\varrho}^{\pi}()] | h_{< t} \right] \quad (12)$$

Where ρ is the environment generating the observations trajectories and ϱ is the predictor with respect to which we calculate point-wise surprise (Def.5.1). $\frac{1}{\Gamma_t}$ serves as a normalization factor, where $\Gamma_t = \sum_{k=t}^{\infty} \gamma_k$ is the sum of all future discounts, and it is often omitted since it does not affect the optimal policy.

Substituting the point-wise surprise $S()$ using Eq. S[o] and expanding the expectation, the full expression becomes

$$\begin{aligned} S_{\rho||\varrho}^{\pi}(h_{< t}) &= \frac{1}{\Gamma_t} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_{\rho}^{\pi}(h_{t:k} | h_{< t}) [-\log \varrho(o_k | h_{< k} a_k)] \end{aligned} \quad (13)$$

Definition 5.3. (The Surprise-Minimizing Agent)

We define the Surprise-Minimizing Agent $\mathbf{sma}(\mathcal{A}, \mathcal{O}, \gamma, \rho, \varrho)$ as the agent which interacts with the world through the cybernetic model (specified in As. 5.3) with action and observation spaces $\mathcal{A}$ and $\mathcal{O}$, respectively. This agent minimizes expected surprise $S_{\rho||\varrho}^{\pi, \gamma}(h_{< t})$ as defined in Eq. 12. Therefore, it acts with optimal policy

$$a_t \sim \pi_{\rho||\varrho}(h_{< t}) \in \arg \min_{\pi} S_{\rho||\varrho}^{\pi, \gamma}(h_{< t}) \quad (14)$$

Definition 5.4. (The Universal Surprise-Minimizing Agent)

We define the Universal Surprise-Minimizing Agent

$$SMA(\mathcal{A}, \mathcal{O}, \gamma, \mathcal{M}, U) := \mathbf{sma}(\mathcal{A}, \mathcal{O}, \gamma, \xi, \xi) \quad (15)$$

as the $\mathbf{sma}(\dots)$ agent, which minimizes expected surprise $S_{\xi||\xi}^{\pi, \gamma}(h_{< t})$ with respect to the universal predictor ξ (described in Sec. 2).

Def. 5.3 is more general than Def. 5.4 as it allows for any pair of environmental models ρ and ϱ .

We write $\mathbf{sma}(\rho||\varrho) := \mathbf{sma}(\cdot, \cdot, \cdot, \rho, \varrho)$ as a shorthand when actions, observations and discounts are implicit. For instance, $\mathbf{sma}(\mu||\xi)$ denotes the μ -optimal "oracle" agent which minimizes the surprise with respect to ξ while knowing the true environment μ . The μ -optimal agent is usually referred to as $AI\mu$ in the literature. On the other hand, the agnostic agent, $AI\xi$ in the literature, in this context can be written as $\mathbf{sma}(\xi||\xi) = SMA(\dots)$. Often, we simply refer to either of these two agents as "SMA".



The reasoning behind this notation is to make the few parameters of the model explicit, similarly to what was done in [Kat19].

for which multiple theorems have already been proven, we could define the reward function as $r_t = \varrho(o_t | h_{< t} a_t)$.

¹³Here we provide the general formulation which includes the discounted horizon, but throughout the text we'll often assume a fixed horizon m for which we set $\gamma_k = \mathbb{I}[k \leq m]$ and write expected cost as $S_{\varrho||\rho}^{\pi, m}$. Nonetheless, we could approximate the infinite discounted horizon by considering instead the effective horizon as defined in [Hut24, Def. 6.4.2].

5.3 Expected Surprise as Entropy

Lemma 5.1. *When $\gamma_k = \llbracket k \leq m \rrbracket$, expected surprise over observations corresponds to the per-step conditional cross-entropy or cross-entropy over trajectories minus the sum of entropy of actions.*

Proof. See Sec. C of the Appendix. $\square$

When $\rho = \varrho$, cross-entropy becomes regular Shannon entropy. Considering cost as surprise on incoming observation only (Eq. S[o]), we can write (Lem. 5.1):

$$S_{\rho \parallel \rho}^{\pi, m}(h_{<t}) = H_{\rho}^{\pi}[\mathcal{H}^{m-t+1} \mid h_{<t}] - \sum_{k=t}^m H_{\rho}^{\pi}[\mathcal{A}_k \mid h_{<t}, \mathcal{H}_{t:k-1}] \quad (16)$$

$$= \sum_{k=t}^m H_{\rho}^{\pi}[\mathcal{O}_k \mid h_{<t}, \mathcal{H}_{t:k-1}, \mathcal{A}_k] \quad (17)$$

We lose the action entropy term when the policy is deterministic or the cost is defined as in Eq. S[h].

Expected surprise over future observations can also be written with the chronological distribution (Sec. 2.5). Considering $\rho \neq \varrho$, we would have:

$$S_{\rho \parallel \varrho}^{\pi, m}(h) = \sum_{h' \in \mathcal{H}^m} P_{\rho}^{\pi}(h' \mid h) [-\log \varrho(o' \parallel a', h)] \quad (18)$$

Where o' and a' are the observations and actions sequences, respectively, within h' , the future interaction sequence. h , instead, is a shorthand for the past interaction sequence $h_{<t}$.

On the other hand, for surprise on the whole action-observation pair (Eq. S[h]), we replace the log term with $[-\log P_{\varrho}^{\pi}(hh')]$, which gives cross-entropy:

$$\begin{aligned} S_{\rho \parallel \varrho}^{\pi, m}(h) &= \sum_{h' \in \mathcal{H}^m} P_{\rho}^{\pi}(h' \mid h) [-\log P_{\varrho}^{\pi}(h' \mid h)] \\ &= H_{\rho \parallel \varrho}^{\pi}[\mathcal{H}^{m-t+1} \mid h_{<t}] \end{aligned} \quad (19)$$

5.4 Motivation

We set the cost to surprise on a single (current) time step, because, as we have seen, expected cost becomes expected surprise over the full trajectory. A quantity on which it has been heavily discussed about in FEP literature. Furthermore, the telescoping properties of the logarithm turn expected surprise into (cross-)entropy of future trajectories (Eq. 19). For this reason, the minimization of $S_{\xi \parallel \xi}^{\pi, m}(h_{<t})$ corresponds to the minimization of motivated by Theorem 4.1(ii). In fact, Eq. 19 corresponds exactly to $\mathbb{E}_{\eta \sim \xi(\eta)} \mathbb{E}_{h \sim P_{\eta}^{\pi}(h)} [-\log P_{\xi}^{\pi}(h)] = \mathbb{E}_{h \sim P_{\xi}^{\pi}(h)} [-\log P_{\xi}^{\pi}(h)]$, but conditioned on the past. At the same time, this definition is consistent with the per-step IG value function defined for KL-KSA, which can be exactly reconstructed from SMA's value, as shown in Equation 21, which is valid for both alternative rewards presented here.

Finally, SMA relates to the predictor model specified in [Hut24, Def. 3.5.1], but extended to the active setting. The main difference consists in the fact that the loss function considered here is unbounded as $\lim_{x \rightarrow 0^+} -\log x = \infty$, while the predictor model considers a bounded loss of 1 when the prediction is incorrect and 0 otherwise.

Nonetheless, when we take the expectation our cost is bounded from above by the dimensionality of the observations (Prop. 5.2). In addition, with Cor. 6.4 we prove that

the bound on expectation over trajectories is half of the bound demonstrated for the predictor model, $2 \log w_{\mu}^{-1}$, demonstrated in [Hut24, Cor. 3.5.14].

5.5 Upper Bounds on Cost and Existence of SMA

This section details the largest range of values our expected cost can take, whose finiteness allows for the existence of an optimal policy. In Sec. 8 we will refine these bounds further for certain types of agents and environments.

Proposition 5.2. *(Co-domain of Expected Surprise)*

Given a summable discount, $\Gamma_t < \infty$, and any class of environments $\mathcal{M}$:

$$0 \leq S_{\xi \parallel \xi}^{\pi, \gamma} \leq \underbrace{\log |\mathcal{O}| \Gamma_t}_{\star} \text{ or } \underbrace{\log |\mathcal{H}| \Gamma_t}_{\diamond} \quad (20)$$

Where the upper bound is $\star$ if we consider cost $S[o]$ or $\diamond$ with $S[h]$.

Proof. The first inequality is motivated in Sec. B.1, while the second follows from Lemma B.1. $\square$

Proposition 5.3. *(Existence of Optimal Policy)* *Given summable discounts $\Gamma_t < \infty$, there exists an optimal policy $\pi_{\xi \parallel \xi}$ as defined in Def. 5.3.*

Proof. This can be demonstrated similarly to what was done for bounded expected reward and discounts in [Lat14, Thm. 10]. In our case, the quantity to be optimized is bounded below by 0, while its expectation is bounded from above (Prop. 5.2). $\square$

These two propositions are inspired by [Ors13, Thm. 2].

6 Surprise as Information Gain plus Ambiguity

In the next section, we outline the relationship between the surprise, minimized by SMA, and information gain, maximized by KSA.

6.1 Decomposition of the Cost Function

Lemma 6.1. *We can decompose expected surprise as information gain plus ambiguity.*

$$S_{\xi||\xi}^{\pi}(h) \stackrel{(a)}{=} \underbrace{\mathbb{E}_{\xi}^{\pi}[IG^{\gamma} | h]}_{\text{information gain}} + \underbrace{\sum_{\nu \in \mathcal{M}} w_{\nu}(h) S_{\nu||\nu}^{\pi, \gamma}(h)}_{\text{ambiguity}} \quad (21)$$

$$\stackrel{(b)}{=} \sum_{\nu \in \mathcal{M}} w_{\nu}(h) KL_h^{\gamma}(P_{\nu}^{\pi} || P_{\xi}^{\pi}) + \sum_{\nu \in \mathcal{M}} w_{\nu}(h) S_{\nu||\nu}^{\pi, \gamma}(h) \quad (22)$$

$$\stackrel{(c)}{=} \sum_{\nu \in \mathcal{M}} w_{\nu}(h) \mathbb{E}_{\nu}^{\pi}[IG_{\nu}^{\gamma} | h] + \sum_{\nu \in \mathcal{M}} w_{\nu}(h) S_{\nu||\nu}^{\pi, \gamma}(h) \quad (23)$$

Proof. See Appendix D for (a). (b) holds because expected IG is equal to the weighted sum of KL divergences between the mixture ξ and each model in $\mathcal{M}$. For a proof, see [Ors13] or Sec. E. In the latter we also provide the formula (Eq. 106) for the KL divergence adapted to the discounted cumulative case adopted in [Ors13]. For equality (c), we used the definition of IG (Eq. 3) and ν -specific IG:

$$IG_{\nu}^{\gamma}(h_{t:\infty}) := \sum_{k=t}^{\infty} \gamma_k \log \frac{w_{\nu}(h_{t:k})}{w_{\nu}(h_{t:k-1})}$$

The full derivation is provided in Appendix E. $\square$

The term *ambiguity* is often used in the FEP literature. It is formulated here as the weighted sum of the surprise in each environment ν , evaluated under the best predictor: ν itself. Therefore, ambiguity represents the stochastic component of the distribution over future observations, as measured by the entropy $H[\nu(\cdot | h_{<t} a_t)]$ (Eq. 17).

This gives the intuition that, under expectation over the ξ mixture itself, SMA should move the posterior to those environments with lowest expected self-surprise $S_{\nu||\nu}^{\pi}$, while limiting the adversarial information gain component.

Ultimately, SMA will experience *true* surprise:

$$S_{\mu||\xi}^{\pi \xi \xi} = \underbrace{KL(P_{\mu}^{\pi \xi \xi} || P_{\xi}^{\pi \xi \xi})}_{\mathbb{E}_{\mu}^{\pi \xi \xi}[IG_{\mu}]} + \underbrace{S_{\mu||\mu}^{\pi \xi \xi}}_{\text{entropy}} \quad (24)$$

This trivially from Lemma 6.1(Eqs. 22 and 23) by placing all weight on μ : $w_{\mu}(h) = 1$. Otherwise, we can interpret $S_{\mu||\xi}^{\pi \xi \xi}$ as cross entropy (Sec. 5.3) and use the known equality which relates KL divergence, entropy and cross entropy:

$$\underbrace{H_h^{\gamma}[P_{\mu}^{\pi} || P_{\xi}^{\pi}]}_{S_{\mu||\xi}^{\pi}(h)} = KL_h^{\gamma}(P_{\mu}^{\pi} || P_{\xi}^{\pi}) + \underbrace{H_h^{\gamma}[P_{\mu}^{\pi} || P_{\mu}^{\pi}]}_{S_{\mu||\mu}^{\pi}} \quad (25)$$

Where H_h^{γ} corresponds to the discounted cumulative entropy with distributions conditioned on h , and is defined analogously to KL_h^{γ} (Eq. 106). Finally, if we consider a model class $\mathcal{M}$ of deterministic environments, the ambiguity component goes

to 0, and we have that $SMA = KMA$ (Tab. 1), as already mentioned.

With this, we are left with μ -specific expected IG or KL divergence as our expected surprise.

Nonetheless, we can equivalently refer to the original definition of $S_{\mu||\xi}^{\pi}$ (Eq. 12), and we will do in the following. To summarize, Lemma 6.1 and Eq. 33 motivates why we will refer to surprise and information gain interchangeably in the coming sections.

6.2 New Upper Bounds

Given what we have seen in the above section, we can provide new upper bounds.

Proposition 6.2. *Models of μ Bound Surprise If $\exists \nu \in \mathcal{M} : P_{\mu}^{\pi}(h' | h) = P_{\nu}^{\pi}(h' | h) \forall h' \in \mathcal{H}^{\infty}$, then:*

$$S_{\mu||\xi}^{\pi} \leq K(\nu) + S_{\mu||\mu}^{\pi} \quad (26)$$

Proof. This follows from Eq. 24 and:

$$\begin{aligned} KL_h^{\gamma}(P_{\mu}^{\pi} || P_{\xi}^{\pi}) &\stackrel{(a)}{\leq} KL_h^{\infty}(P_{\mu}^{\pi} || P_{\xi}^{\pi}) \\ &\stackrel{(b)}{=} \sum_{h_{t:\infty} \in \mathcal{H}^{\infty}} P_{\mu}^{\pi}(h_{t:\infty} | h_{<t}) \log \frac{P_{\mu}^{\pi}(h_{t:\infty} | h_{<t})}{P_{\xi}^{\pi}(h_{t:\infty} | h_{<t})} \\ &\leq \sum_{h_{t:\infty} \in \mathcal{H}^{\infty}} P_{\mu}^{\pi}(h_{t:\infty} | h_{<t}) \log \frac{P_{\mu}^{\pi}(h_{t:\infty} | h_{<t})}{w_{\nu} P_{\nu}^{\pi}(h_{t:\infty} | h_{<t})} \\ &\stackrel{(c)}{=} \sum_{h_{t:\infty} \in \mathcal{H}^{\infty}} P_{\mu}^{\pi}(h_{t:\infty} | h_{<t}) \log \frac{1}{w_{\nu}} \stackrel{(d)}{\leq} K(\nu) \end{aligned}$$

(a) because $\gamma \leq 1$ and ∞ represents the discount $\gamma_t = 1 \forall t$, while (b) due to the known telescoping property of KL (Sec. E.1). (c) is true by the hypothesis of the proposition. (d) holds because P_{μ}^{π} is a (semi)measure and $\log \frac{1}{w_{\nu}} = K(\nu)$. $\square$

Without requiring the condition of the above proposition, we can also provide a general bound with the following.

Proposition 6.3.

$$S_{\mu||\xi}^{\pi} \leq \inf_{\nu \in \mathcal{M}} \{K(\nu) + KL_h^{\infty}(P_{\mu}^{\pi} || P_{\nu}^{\pi})\} + S_{\mu||\mu}^{\pi} \quad (27)$$

Proof. This is proven as Prop. 6.2, the additional KL term appears because $P_{\mu}^{\pi}(h' | h) = P_{\nu}^{\pi}(h' | h)$ is not necessarily respected. $\square$

The above is based on [Hut24, Thm. 3.4] and is especially useful in the case where $\mu \notin \mathcal{M}$. Otherwise, the following bound applies as well.

Corollary 6.4. *If $\mu \in \mathcal{M}$:*

$$S_{\mu||\xi}^{\pi} \leq K(\mu) + S_{\mu||\mu}^{\pi} \quad (28)$$

Proof. μ trivially satisfies the condition of Prop. 6.2. $\square$

This bound is not necessarily tighter than the bound shown in Prop. 5.2.

7 The Oracle Agent

In this section, we discuss how the oracle agent $\mathbf{sma}(\mu||\xi)$, knowing μ , fulfills its objective, the properties of the latter, and under which conditions SMA explores its environment. We analyze the extent of the exploration, quantified by the amount of knowledge gathered, despite it being against the objective of surprise minimization in the short term. We focus on the oracle agent, which acts with μ -optimal policy $\pi_{\mu||\xi}$ (Def. 5.3), because it provides a lower bound on the true surprise experienced by the agnostic agent $\mathbf{sma}(\rho||\rho)$, acting with policy $\pi_{\xi||\xi}$. In fact, $S_{\mu||\xi}^{\pi_{\mu||\xi}} \leq S_{\mu||\xi}^{\pi_{\xi||\xi}}$ holds by definition of optimal policy.

We will generally consider a finite horizon $m \in \mathbb{N}^+$, with discounts $\gamma_k = [[k \leq m]]$, which can be thought of as the expected lifetime of the agent. As shown in Sec. 5.3, this allows for a cleaner interpretation of the expected surprise, and will allow us to equate it to the complexity of future observations (Sec. 7.4).

7.1 Fitting a Niche of the Environment

Given what we have just seen with Eqs. 21 and 24, intuition suggests that such an agent would look for the closest subsection of its environment which fits with prior beliefs, leading to the lowest cumulative Information gain or KL-divergence. That is, it would end up in the closest sub-environment μ' with the lowest Kolmogorov complexity, as ξ has a prior on simplicity, and would perform non-exploratory actions to maintain the beliefs of ξ on such μ' . We would generally have $K(\mu') \leq K(\mu)$. Where equality holds if ξ excludes all the environments with complexity lower than $K(\mu)$.

We formally define the niche μ' of an environment μ as follows.

Definition 7.1 (π -niche). An environment μ' , belonging to the class of deterministic environments $\mathcal{M}_{det}$ ($\mathcal{M}$ for short) is a π -niche of an environment $\mu \in \mathcal{M}$, if it has lower complexity than μ and gives the same observations under policy π :

$$\begin{aligned} \mu' &\in \mathcal{M}_{\leq \mu}^{\pi, m}(h) \\ &:= \{\nu \in \mathcal{M} | K(\nu) \leq K(\mu), \exists h' \in \mathcal{H}^m : P_{\nu}^{\pi}(hh') = P_{\mu}^{\pi}(hh')\} \end{aligned} \quad (29)$$

With $h' \in \mathcal{H}^{m-|h|}$. More generally, we say that μ' is a niche of μ iff: $\exists \pi : \mu' \in \mathcal{M}_{\leq \mu}^{\pi, m}(h)$

This is the converse of h-separability, defined in [Ors14], which defines two deterministic environments "separable" when there is a policy that can distinguish them (given history h and horizon m).

By changing the inequality, we can select environments with higher complexity $\mathcal{M}_{\geq \mu}^{\pi, m}(h)$. Instead, when none is specified, we get all the environments consistent with μ : $\mathcal{M}_{\mu}^{\pi, m}(h)$. We could also define a specific complexity threshold k with and additional subscript, e.g. $\mathcal{M}_{\mu, \geq k}^{\pi, m}(h)$.

Definition 7.2. (Sum of Weights of Consistent Environments) The sum of the weights belonging to the selected set, $\mathcal{M}_{\mu}^{(\dots)}(\dots)$, is defined with the following notation:

$$\mathbf{W}_{\leq \mu}^{\pi, m}(h) = \sum_{\nu \in \mathcal{M}_{\leq \mu}^{\pi, m}(h)} w_{\nu}$$

Applying this definition we obtain the following equality.

Proposition 7.1 (Re-writing cost). Assuming a deterministic policy and h sampled from μ :

$$S_{\mu||\xi}^{\pi, m}(h) = \log [\mathbf{W}_{\mu}^0(h)] - \log [\mathbf{W}_{\mu}^{\pi, m}(h)] \quad (30)$$

Proof.

$$S_{\mu||\xi}^{\pi, m}(h) \quad (31)$$

$$\stackrel{(a)}{=} \sum_{h' \in \mathcal{H}^m} P_{\mu}^{\pi}(h' | h) \left[-\log \frac{\xi(o'o||aa')}{\xi(o||a)} \right] \quad (32)$$

$$\begin{aligned} &= \sum_{h' \in \mathcal{H}^m} P_{\mu}^{\pi}(h' | h) \left[-\log \frac{\sum_{\nu \in \mathcal{M}} w_{\nu} \nu(o'o||aa')}{\sum_{\nu \in \mathcal{M}} w_{\nu} \nu(o||a)} \right] \\ &\stackrel{(b)}{=} -\log \left[\frac{\sum_{\nu \in \mathcal{M}_{\mu}^{\pi, m}(h)} w_{\nu}}{\sum_{\nu \in \mathcal{M}_{\mu}^{\pi, 0}(h)} w_{\nu}} \right] = \log [\mathbf{W}_{\mu}^0(h)] - \log [\mathbf{W}_{\mu}^{\pi, m}(h)] \end{aligned} \quad (33)$$

(a) since Eq. 18 applies to both definitions of cost when the policy is deterministic. (b) because $P_{\mu}^{\pi}(h' | h)$ selects the unique history h' , which gives $P_{\mu}^{\pi}(h' | h) = 1$, coherent win environment μ and policy π , and, on the numerator $\nu(o'o||aa') = 1$ selects those environments ν with $P_{\nu}^{\pi}(hh') = 1$, while on the denominator, $\nu(o||a) = 1$ selects the weights of the environments consistent with history h . $\square$

Lemma 7.2. Given an empty history, $h = \epsilon$, the expected surprise given by the true environment is inversely proportional to the sum of the weights of the environments, scaled by discounts, which are kept consistent with μ up to horizon m . When $\gamma_t = [[t \leq m]]$, we can write:

$$S_{\mu||\xi}^{\pi, m}(\epsilon) = -\log [\mathbf{W}_{\mu}^{\pi, m}(\epsilon)] \quad (34)$$

Proof. Can be proven as Prop. 7.1, or using the result from the latter and the fact that $\log [\mathbf{W}_{\mu}^0(\epsilon)] \approx 0$ $\square$

Altogether, this gives an ordering on the expected cost of different policies based on the amount of prior they manage to preserve. In fact, since $\log [\mathbf{W}_{\mu}^0(h)]$ doesn't depend on the (future) policy:

$$S_{\mu||\xi}^{\pi, m}(h) < S_{\mu||\xi}^{\pi', m}(h) \Leftrightarrow \mathbf{W}_{\mu}^{\pi, m}(h) > \mathbf{W}_{\mu}^{\pi', m}(h) \quad (35)$$

With a regular discount function, the importance given to discarded environments at time t decreases with the discount factor γ_t . As we will see, it can get increasingly harder for SMA to keep environments consistent for a longer (effective) horizon.

7.2 Leaving The Niche (Intuition)

Even knowing μ , $\mathbf{sma}(\mu||\xi)$ would have to pick the deterministic action sequence which leads to the lowest amount of environments discarded by the agnostic predictor ξ , weighted by their prior.

This is the opposite of what KSA would do in deterministic environments as explained in [Ors14]. For this reason, the latter tends to discard simpler models first, while SMA would prefer to discard more complex environments as they have a smaller impact on the sum of priors. The author argues it could allow the agent to converge faster to the true environment. Nonetheless, we believe this could cause learning stability issues in bounded agents.

In fact, in the initial phase of the agent's life, KSA will rapidly discard models with low complexity. These are precisely the ones with higher weight in the mixture ξ . As a result, in this early stage, the agent will erratically alter its beliefs about future time steps. Furthermore, if the true environment is complex enough, the agent's memory would not be able to store it in its entirety and would indeed be motivated to look for a stable niche with a shorter description.

In this subsection, we leverage Lem. 7.2 to explore the case in which SMA behaves similarly to KSA, by choosing a policy π that discards the simplest environment μ' , which could technically fit μ :

$$\mu' := \arg \inf_{\nu \in \mathcal{M}_\mu^m(h)} K(\nu), \quad \pi' = \arg \sup_{\pi} \mathbf{W}_{\mu,=\mu'}^{\pi,m}(h) \quad (36)$$

We have:

$$S_{\mu||\xi}^{\pi_{\mu||\xi},m} \stackrel{(a)}{\leq} S_{\mu||\xi}^{\pi',m} \stackrel{(b)}{\leq} K(\mu') \quad (37)$$

(a) holds by definition of optimal policy, while (b) follows from Prop. 6.2.

μ' will be discarded by SMA if it is necessary to maintain a bigger portion of complex environments, which altogether might have an higher total prior.

In fact, if we have a policy π which discards $\mathcal{M}_{\mu,=\mu'}^{\pi',m}$ and another policy π' that doesn't, due to Eq. 35 $\mathbf{sma}(\mu||\xi)$ would still prefer π if:

$$\begin{aligned} \mathbf{W}_{\mu}^{\pi,m}(h) &> \mathbf{W}_{\mu}^{\pi',m}(h) \\ \mathbf{W}_{\mu,>\mu'}^{\pi,m}(h) &> \mathbf{W}_{\mu,=\mu'}^{\pi',m}(h) + \mathbf{W}_{\mu,>\mu'}^{\pi',m}(h) \end{aligned} \quad (38)$$

This intuitively happens with decreasing probability with smaller $K(\mu')$ and consequently higher $\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h)$.

In fact, since prior weights have to sum up to a value ≤ 1 , $\mathbf{W}_{\mu,>\mu'}^{\pi,m}(h)$ is bounded by:

$$\mathbf{W}_{\mu,>\mu'}^{\pi,m}(h) \leq 1 - \mathbf{W}_{<\mu'} - \mathbf{W}_{=\mu'} \leq 1 - \mathbf{W}_{<\mu'} - \mathbf{W}_{\mu,=\mu'}^{\pi',m} \quad (39)$$

Where $\mathbf{W}_{<\mu'}$ is the prior weight of envs. with complexity $< \mu'$, these have either already been discarded by the observation history or they cannot be consistent with environment μ up to horizon m for any policy. If they were, one would have been selected as the lowest complexity niche μ' by Eq. 36.

Theorem 7.3 (Staying in the niche). *If there is a policy π' such that it can keep the simplest niche μ' (Eq. 36) consistent with μ up to horizon m and:*

$$\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h) \geq \frac{1 - \mathbf{W}_{<\mu'}}{2}$$

$\mathbf{sma}(\mu||\xi)$ will surely keep μ' consistent with μ :

$$\mu' \in \mathcal{M}_{\mu||\xi}^{\pi_{\mu||\xi},m}(h) \quad (i)$$

The obtained IG_μ and surprise are bounded by $K(\mu')$.

$$S_{\mu||\xi}^{\pi_{\mu||\xi}} = \mathbb{E}_{\mu||\xi}^{\pi_{\mu||\xi}}[IG_\mu] \leq K(\mu') \quad (ii)$$

Proof. Statement (i) stems from the fact that $\mathbf{W}_{\mu,>\mu'}^{\pi,m}(h)$ is bounded by the remaining available weight (Eq. 39). Consequently, the inequality in Eq. 38 would surely not hold if:

$$\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h) \geq 1 - \mathbf{W}_{<\mu'} - \mathbf{W}_{=\mu'} \quad (41)$$

$$\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h) + \mathbf{W}_{=\mu'} \geq 1 - \mathbf{W}_{<\mu'} \quad (42)$$

A sufficient condition for the above to hold is:

$$2\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h) \geq 1 - \mathbf{W}_{<\mu'} \quad (43)$$

$$\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h) \geq \frac{1 - \mathbf{W}_{<\mu'}}{2} \quad (44)$$

Where for Eq. 43 we used the fact that by definition $\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h) \leq \mathbf{W}_{=\mu'}$.

(ii) follows immediately from Lemma 7.2 and $\mathbf{W}_{\mu||\xi}^{\pi_{\mu||\xi},m}(h) \geq \mathbf{W}_{=\mu'}^{\pi_{\mu||\xi},m}(h) \geq 2^{-K(\mu')}$ $\square$

The intuition for Eq. 44 is that $\mathbf{W}_{\mu,=\mu'}^{\pi',m}(h)$ needs to take up only half of the remaining available weight, $1 - \mathbf{W}_{<\mu'}$, in order to be greater or equal to $\mathbf{W}_{\mu,>\mu'}^{\pi,m}(h)$, which can only take up the remaining part (weights sum to ≤ 1).

Given $N_{\mu'}^{\pi'} = N_{\mu'}^{\pi',m} := |\mathcal{M}_{\mu,=\mu'}^{\pi',m}(h)|$, the number of π' -niches with same complexity as μ' , we can also re-write Eq. 43 as:

$$N_{\mu'}^{\pi'} 2^{-K(\mu')+1} \geq 1 - \mathbf{W}_{<\mu'} \quad (45)$$

Example 7.1 (Really simple niche). Let's consider a case where the simplest niches, that have not yet been discarded by h , have a complexity of $K(\mu') = 3$ and $N_3 = 2$. This means that the Kolmogorov complexity given by UTM U generates a prefix code that has two programs with a complexity of 3 and π' manages to keep both of those consistent with observations up to horizon m , so that $N_{\mu'} = |\mathcal{M}_{\mu,=\mu'}^{\pi',m}(h)| = 2$. Then, if the discarded environments so far (considering also the ones that cannot be made consistent up to horizon m) are two environments with complexity = 2, so that

$$\mathbf{W}_{<\mu'} = N_2 \cdot 2^{-k} = 2 \cdot 2^{-2} = \frac{1}{2}$$

, it follows that Eq. 45 holds since:

$$N_{\mu'}^{\pi'} 2^{-K(\mu')+1} = 2 \cdot 2^{-3+1} = 1 - \mathbf{W}_{<\mu'} = \frac{1}{2}$$

Hence, no policy π which discards all environments with complexity = $K(\mu')$ will be preferred to π' .

 Interestingly, as this example hints, we don not need to verify if all the (possibly infinite environments) fit with μ or not to decide on a "good enough" policy. In fact, we could make sure that only a single (or a few if we are uncertain about μ) simple environment fits with the true environment, since this already bounds our cost (Eq. 37).

7.3 Niches are Minimal Programs

As we will see, it is hard, if not impossible, to give general exploration/surprise bounds, other than those presented in Sec. 6.2 and Prop. 5.2. Notwithstanding, it is important to mention that in what we may consider a "natural" setting¹⁴, there are no niches which have significantly lower complexity than the

¹⁴In [Hut07, Sec. 2.8] the "natural" setting considered is an agent with $1 \ll m \ll |\mathcal{O}| \approx |\mathcal{A}|$, as it takes a live video as input observations and outputs a video to monitor for the actions. On the other hand, we additionally require $|\mathcal{A}| \ll |\mathcal{O}|$ to provide meaningful bounds.

true environment μ^{15} . We demonstrate this in Sec. 8.1, but we first need the following lemmas and theorems to understand the behavior of $\mathbf{sma}(\mu||\xi)$ in a generic environment.

Lemma 7.4. *For any given policy, the complexity of the simplest niche is equal, up to a constant, to the complexity of the observation sequence conditioned on the action sequence. In general:*

$$\forall a : K(\mu'_a) \stackrel{\pm}{=} K(o | a) \quad \text{with } \mu(a) = o \quad (46)$$

Where $\mu'_a \in \mathcal{M}_{\mu}^{\pi, m}$ is the simplest π -niche for μ , given action sequence $a = a_{1:m} \sim \pi$ and horizon m . $o = o_{1:m}$ is a shorthand.

Proof. Proving $K(\mu'_a) \stackrel{\pm}{=} K(o | a)$:

Firstly, we write down the following.

$$K(\mu'_a) \stackrel{(a)}{\leq} K(\nu_{p_a^*}) \stackrel{(b)}{=} K(p_a^*) \stackrel{+}{\leq} l(p_a^*) = K(o | a) \quad (47)$$

p_a^* is the minimal program such that $l(p_a^*) = K(o | a)$, with $o = \mu(a)$. (a) holds by definition of minimal niche, since $\nu_{p_a^*} \in \mathcal{M}_{det}$ is the environment associated with p_a^* . Since programs enumerate environments, (b) is true (Sec. 2.4). The third relationship is well known and applies to minimal programs [Li19, Example 3.2.3]¹⁶. The fourth follows by definition of minimal program p_a^* .

Proving $K(\mu'_a) \stackrel{+}{\geq} K(o | a)$:

Consider the program p associated with environment μ'_a , which has the property $U(p, a) = o$. The following holds.

$$K(o | a) \stackrel{+}{\leq} K_V(o | a) \stackrel{(a)}{\leq} l(q) = K(p) = K(\mu'_a) \quad (48)$$

The first inequality follows from the invariance theorem. For the second inequality, we define V as a TM which recovers p and then runs it on U to output o given a . More precisely, we can write $V(q) = U(U(q), a) = U(p, a)$. Where q is a minimal program which outputs p : $l(q) = K(p)$. Since q is a program which runs on V and returns o , given side information a , (a) follows by definition of Kolmogorov complexity (Sec. 2.4). $\square$

While the above lemma holds for any policy, the following is valid for the one which assures the simplest niche possible.

Lemma 7.5 (Niches are minimal programs). *The complexity of the simplest niche μ' (Eq. 36) is equal, up to a constant, to the length of the minimal program that produces the observation sequence $o = o_{1:m} \sim \mu(\cdot||a_{1:m})$ under the optimal action sequence. $a_{1:m} = a$ is a shorthand.*

$$K(\mu') \stackrel{\pm}{=} \min_a K(o|a) \quad \text{with } o = \mu(a) \quad (49)$$

¹⁵We can interpret μ as the true “local” environment in space and time; meaning the environment which can fully represent, off-policy, the true environment up to horizon m and given history $h_{<t}$. Note, this is different from the minimal niche which can represent the true environment only on-policy. Nonetheless, the properties presented below hold both if we consider μ to be locally true or absolutely true.

¹⁶In general, given a minimal program p^*

$$K(p^*) \leq l(p^*) + O(1)$$

In fact, we can modify U with a constant overhead i to create an identity machine $U(i|p^*) = I(p^*) = p^*$ with $p^* \in L(U)$. For example, this could be done by defining I as U with a slightly different transition function and few more states so that it can effectively simulate U . The accepting states can then be replaced with a state that triggers the copying of the input tape from current position of the reading head to the start, making sure the head is put back into its final position, and then transitions to an accepting state once it has fully transcribed the input program.

Proof. Proving $K(\mu') \stackrel{+}{\leq} \min_a K(o|a)$:

We name a^* the action sequence such that:

$$a^* = \arg \min_a K(o | a) \quad \text{with } o = \mu(a)$$

Then, $K(\mu') \leq K(\mu'_{a^*})$ holds by definition (Eq. 36), and $K(\mu'_{a^*}) \stackrel{+}{=} \min_a K(o | a)$ by Lem. 7.4.

Proving $K(\mu') \stackrel{+}{\geq} \min_a K(o|a)$:

μ' by definition produces the same observations as μ under policy π' . Then, we can deterministically get the actions and observations sequences a', o' within history: $h' \sim P_{\nu'}^{\pi'}(h' | h)$.

Finally, $K(o' | a') \stackrel{+}{\leq} K(\mu')$ holds for the same reasoning adopted for Eq. 48, while $\min_a K(o | a) \leq K(o' | a')$ is trivially true. $\square$

The intuition behind Lemma 7.5 is that searching for the simplest niche that can simulate μ up to horizon m means, from the perspective of the $\mathbf{sma}(\mu||\xi)$ agent which knows μ , looking for the action sequence which minimizes the conditional complexity of the observation sequence.

7.4 Behavior of The Oracle Agent

In the previous section, we have shown how the simplest niche possible μ' relates to $K(o | a)$ (Lemma 7.5).

In this section, we reason further on whether $\mathbf{sma}(\mu||\xi)$ keeps the simplest niche μ'^{17} or if it would manage to receive a lower cost by focusing on keeping more complex environments, rather than few simple ones, consistent with μ . This is what we already discussed in Section 7.2 from the perspective of the sum of weights of a prefix-free set such as the ones given by K with Solomonoff prior.

In order to do this, we characterize $\mathbf{sma}(\mu||\xi)$'s policy by equating its cost function to $K(o | a)$ leveraging the coding theorem, which entails that our agent will try to minimize the complexity of the observations stream. Intuitively, according to the coding theorem, if a sequence o , has many long descriptions, it must have a short description too and the shortest dominates the sum of prior weights [Li19, Example 4.3.5]. More formally:

Theorem 7.6 (Expected Surprise as Complexity). *Given a deterministic policy π (Sec.2.7):*

$$S_{\mu||\xi}^{\pi, m}(\epsilon) \stackrel{\pm}{=} K(o|a)$$

with

$$o, a \text{ in: } h \sim P_{\mu}^{\pi}(\cdot)$$

Proof. We start from the fact that:

$$S_{\mu||\xi}^{\pi, m}(\epsilon) = -\log \xi(o||a)$$

where o and a are deterministically generated by $P_{\mu}^{\pi}(\cdot)$. Then we demonstrate:

$$\log \xi(o||a) \stackrel{\pm}{=} \log M(o||a) \stackrel{\pm}{=} K(o | a) \quad (50)$$

The above equalities are based on the conditional coding theorem [Li19, Thm. 4.3.4]. Nonetheless, the latter consider model class $\mathcal{M}_{sol}$ of lower semi-computable semi-measures, therefore we adapt it for the AIXI setting and the deterministic environments $\mathcal{M}_{det}$ considered here. For the full proof see Appendix F. $\square$

¹⁷Or other environments with the same complexity.

Corollary 7.7 (Universal Surprise-minimizing Agents Seek Simple Niches).

$$\min_{\pi} S_{\mu||\xi}^{\pi, m} = S_{\mu||\xi}^{\pi_{\mu||\xi}, m} \stackrel{\pm}{=} \min_a K(o | a) \stackrel{\pm}{=} K(\mu')$$

This means that we can obtain a c -optimal policy by minimizing $K(o | a)$ (or even $M(o||a)$), instead of $S_{\mu||\xi}^{\pi, m}$.

Proof. The first equality follows from the definition of optimal policy $\pi_{\mu||\xi}$. For the second equality we prove $\stackrel{\pm}{=}$ by fixing o', a' in: $h \sim P_{\mu||\xi}^{\pi_{\mu||\xi}}(\cdot)$. Then we have:

$$S_{\mu||\xi}^{\pi_{\mu||\xi}, m} = \xi(o' || a') \stackrel{\pm}{=} K(o' | a') \geq \min_a K(o | a)$$

We can prove $\stackrel{\pm}{=}$ with a similar reasoning. Finally, the third equality follows from Lemma 7.5. $\square$

Altogether Lemma 7.5, Theorem 8.1 and Cor. 7.7 allow us to characterize the behavior of the **sma**($\mu||\xi$) agent almost fully.

Finally, the expected cost, with a given history h , can be formulated with the following:

Corollary 7.8. *Given an interaction history h coherent with (or sampled from) μ :*

$$S_{\mu||\xi}^{\pi, m}(h) \stackrel{\pm}{=} K(o o' | a a') - K(o | a)$$

with

$$o, a \text{ in } h \quad \text{and} \quad o', a' \text{ in } h' \sim P_{\mu}^{\pi}(h' | h)$$

Proof. Given o, o', a and a' defined as above, we can write:

$$\begin{aligned} S_{\mu||\xi}^{\pi, m}(h) &= -\log \frac{\xi(o o' || a a')}{\xi(o || a)} \\ &= -\log \xi(o o' || a a') - [-\log \xi(o || a)] \end{aligned}$$

and proceed by equating these two terms to $K(o o' || a a')$ and $K(o | a)$ respectively, as done in Thm. 7.6. $\square$

In the following sections we will generally consider an empty history for simplicity.

8 Escaping the Dark Room Problem

Given what we have seen so far, it seems the oracle agent would fall into the dark room problem. Raised by critics of the free energy principle, it states that if biological systems were to avoid surprises “they would simply seek a dark, unchanging chamber, and stay there”[Fri12]. Firstly, we can assume that a dark room corresponds to a low complexity environment within the agent’s model pool $\mathcal{M}$. For instance, consider one which always outputs the same pitch black image¹⁸ as observation. Without any further assumptions about μ , there could easily be a specific action sequence a^* which outputs a low complexity observation sequence o , with $K(\mu') \stackrel{\pm}{=} \min_a K(\mu(a) | a) = K(o | a^*) \stackrel{\pm}{=} K(o) \stackrel{\pm}{=} O(1)$, such as the one we could find in a dark room. If that were the case, **sma**($\mu||\xi$) would indeed select this action sequence leading to minimal surprise (Cor. 7.7) and information gain.

¹⁸This can be thought of as a stream of zeros corresponding to the concatenated RGB values of the pixels in the image, when encoded in binary digits on the output tape.

8.1 Incompressibility of Environments

To resolve the dark room problem, the following theorem establishes a lower bound on the complexity of the simplest niche by leveraging a specific set of “naturalistic” assumptions regarding the sensory-action interface and the environment.

Theorem 8.1 (Incompressibility of minimal environments). *We write the dimensionality of the observation and action sequences in bits $l(o) = \log |\mathcal{O}^m| = m \log |\mathcal{O}|$ and $l(a) = \log |\mathcal{A}^m| = m \log |\mathcal{A}|$ respectively.*

Now, assume:

As. I. *Observations have a much higher dimensionality than action sequences:*

$$l(a) \ll l(o)$$

As. II. *The true environment is far more complex than the dimensionality of the action sequences*

$$l(a) \ll K(\mu)$$

As. III. *Observations provide enough (above a certain constant) information about μ :*

$$\forall a \in \mathcal{A}^m : K(\mu | o, K(o | a), a) \leq c_{\mu} < K(\mu)$$

Where c_{μ} is a constant which depends only on μ . Then we can prove bounds on the complexity of the smallest possible niche μ' (Eq. 36):

$$K(\mu) - l(a) - 2 \log l(a) - c_{\mu} \stackrel{\pm}{=} K(\mu') \quad (51)$$

Proof. Using the conditional symmetry of information theorem [Li19, Thm. 3.8.1 and Eq. 3.22] we have that:

$$\begin{aligned} K(\mu | a) + K(o | \mu, K(\mu | a), a) &\stackrel{\pm}{=} K(o | a) + K(\mu | o, K(o | a), a) \\ K(\mu | a) + O(1) &\stackrel{\pm}{=} K(o | a) + K(\mu | o, K(o | a), a) \end{aligned}$$

In fact, $K(o | \mu, K(\mu | a), a) = O(1)$ holds because we can recover o from μ and a , with no additional information. Absorbing $O(1)$ in $\stackrel{\pm}{=}$ and moving one term to the left:

$$I(\mu; o | a) \stackrel{(a)}{=} K(\mu | a) - K(\mu | o, K(o | a), a) \stackrel{\pm}{=} K(o | a) \quad (52)$$

(a) holds by definition of conditional mutual information in [Li19, Eq. 3.21] Intuitively, this states that the amount of information about μ contained in o is equal to all the information (or complexity) contained in o itself.¹⁹ Now, applying the regular symmetry of information theorem to the (μ, a) pair we obtain:

$$\begin{aligned} K(\mu | a, K(a)) + K(a) &\stackrel{\pm}{=} K(\mu, a) \stackrel{\pm}{=} K(a | \mu, K(\mu)) + K(\mu) \\ K(\mu | a, K(a)) &\stackrel{\pm}{=} K(a | \mu, K(\mu)) + K(\mu) - K(a) \end{aligned} \quad (53)$$

we can write the following, considering the well known information non-increase of conditional complexity [Hut24, Thm. 2.7.15], $K(\mu | a) \stackrel{\pm}{=} K(\mu | a, K(a))$, together with Eq. 53.

$$K(\mu | a) \stackrel{\pm}{=} K(a | \mu, K(\mu)) + K(\mu) - K(a) \geq K(\mu) - K(a) \quad (54)$$

¹⁹This definition of mutual information can be equated with expected information gain which fits perfectly with the story $S_{\mu||\xi}^{\pi} = \mathbb{E}_{\mu}^{\pi}[IG_{\mu}]$ (Eq. 24) considering that, as we will see later, $K(o | a)$ equates to expected surprise $S_{\mu||\xi}^{\pi}$.

Now, since the value $K(a)$ is bounded from above [Hut24, Thm. 2.7.11] as $K(a) \stackrel{\pm}{\leq} l(a) + 2 \log l(a)$ we obtain:

$$K(\mu | a) \stackrel{\pm}{\underset{(b)}{\geq}} K(\mu) - K(a) \stackrel{\pm}{\geq} K(\mu) - l(a) - 2 \log l(a) \quad (55)$$

Where (b) holds by Eq. 54. Plugging the above inequality in Eq. 52 we obtain:

$$K(\mu) - l(a) - 2 \log l(a) - K(\mu | o, K(o | a), a) \leq K(o | a) \quad (56)$$

Finally, using As. III we have that:

$$\forall a : K(\mu) - l(a) - 2 \log l(a) - c_\mu \leq K(o | a) \quad (57)$$

which means:

$$K(\mu) - l(a) - 2 \log l(a) - c_\mu \leq \min_a K(o | a) \stackrel{\pm}{\leq} K(\mu') \quad (58)$$

Where the last equality follows from Lemma 7.5. $\square$

Conjecture 8.1. The lower bound presented in Thm. 8.1, here named “LB”, is tight.

$$K(\mu') - LB \stackrel{\pm}{\leq} l(a) + 2 \log l(a)$$

Where:

$$LB = K(\mu) - l(a) - 2 \log l(a) - c_\mu \leq K(\mu | a) - c_\mu \stackrel{\pm}{\leq} K(\mu') \quad (59)$$

Proof. See Appendix H. $\square$

We would trivialize the bound in Thm. 8.1 with $\mu = \nu_{p^*}$, such that:

$$K(\mu) = K(\nu_{p^*}) \stackrel{\pm}{\leq} l(p^*) = \min_a K(o | a), \quad \text{where: } o = \mu(a) \quad (60)$$

Which entails $\mu' = \mu$. Nonetheless, we are interested in the general case where the true environment is compressible with $K(\mu') < K(\mu)$. This could be easily obtained with the true environment $\mu = \nu_{o^*}$ which ignores the action sequence, and always generates the same observations o with minimal program o^* and $K(\mu) = K(\nu_{o^*}) \stackrel{\pm}{\leq} l(o^*) = K(o)$. With this we have that the simplest niche μ' will instead take advantage of all bits in the action sequence to generate o with a shorter program length, in fact $\min_a K(o | a) \stackrel{\pm}{\leq} K(o)$.

Nonetheless, a much greater subset of environments, have observations which depend on actions. Again, it is hard, in general, to provide a lower bound of the observation sequence given any arbitrary input actions, and to encode an action-dependent environment which respects As. III. We achieve this in the next section.

8.2 Action-dependent Compression

The following lemma provides a construction that justifies As. III by considering environments which are formed by an action-dependent and action-independent component. The latter is represented as a minimal environment which generates observations which cannot be compressed further, even though this could be generalized to environments with redundant and compressible information in their description, representing a constant overhead which could be added to c_μ . The latter quantifies the magnitude of missing information about μ that we cannot gather from the observations, which are adversarially least informative (Cor. 7.7).

Lemma 8.2 (Upper bound on action-dependent compression). *There are environments for which there exist a constant c_μ such that:*

$$\forall a \in \mathcal{A}^m, o = \mu(a) : K(\mu | o, K(o | a), a) \stackrel{\pm}{\leq} c_\mu \ll K(\mu)$$

Proof. Consider an environment μ described by a tuple $\langle T_{i+}, \mu^*, T_j, \rangle$.

T_{i+} takes as input the actions sequence and two Turing machines T_j and μ^* , runs them both separately, and adds their outputs at every time step such that:

$$U(i^+ \mu^* j, a) = T_{i+}(\mu^* j, a) = o_{\mu^*} + o_j = o = \mu(a) \quad (61)$$

Where $o_j = T_j(a)$ and the operation of addition between observation sequences is defined as follows for each time step t .

$$o_t = (o_{t, \mu^*} + o_{t, j}) \pmod{|\mathcal{O}|} \quad (62)$$

Importantly, while the output of T_j depends on the action, μ^* ignores them and always produces the same observation sequence o_{μ^*} for which μ^* is the minimal program:

$$K(o_{\mu^*} | a) \stackrel{\pm}{\leq} K(o_{\mu^*}) = l(\mu^*)$$

With this, we can reconstruct μ knowing the interaction sequence (o and a) and the action-dependent component T_j and T_{i-} . In fact, we are only missing μ^* which can be obtained with a constant amount of bits making use of the interaction sequence as:

$$T_{i-}(o' j, a) = o - o_j \pmod{|\mathcal{O}|} = o_{\mu^*} \quad (63)$$

Where the difference between action sequences is defined analogously as the addition (Eq. 62). Now we have everything to reconstruct μ^* :

$$K(\mu^* | o, K(o | a), a) \stackrel{\pm}{\leq} K(\mu^* | o_{\mu^*}, K(o_{\mu^*})) + l(i^- j) + l(b') \quad (64)$$

The $l(i^- j)$ term serves to compute o_{μ^*} from o as mentioned above (Eq. 63), while the $l(b')$ term is used to calculate $K(o_{\mu^*})$ from $K(o | a)$. In fact, $|K(o | a) - K(o_{\mu^*})| \leq 2l(i^- j) + K(a)$ because of the following inequalities given by construction:

$$K(o | a) \stackrel{\pm}{\leq} K(o_{\mu^*}) + l(i^+) + l(j) \quad (65)$$

$$K(o_{\mu^*}) - K(a) \leq K(o_{\mu^*} | a) \stackrel{\pm}{\leq} K(o | a) + l(i^-) + l(j) \quad (66)$$

$b \in \mathbb{N}^0$ is a number such that $K(o | a) + b = K(o_{\mu^*})$, with $b \leq 2(l(i^- j)) + K(a) \leq 2(l(i^- j)) + l(a) + 2 \log l(a)$ and:

$$\begin{aligned} l(b') &\leq \log [2l(i^- j) + l(a) + 2 \log l(a)] \\ &\quad + \log \log [2l(i^- j) + l(a) + 2 \log l(a)] \\ &= l(l(2l(i^- j) + l(a) + 2 \log l(a)))' \\ &\leq l(l(l(i^- j) + l(a)))' + 2 \end{aligned} \quad (67)$$

Moreover, in Eq. 64 the term $K(\mu^* | o_{\mu^*}, K(o_{\mu^*})) \stackrel{\pm}{\leq} O(1)$, since, in general, $K(x^* | x, K(x)) \stackrel{\pm}{\leq} O(1)$ [Gác21, pg. 67]. We can call r the TM program which computes $T_r(b' T_{i-}(j, o, a), K(o | a)) = \mu^*$ by dove-tailing all programs with minimal length $K(o | a) + b$ enumerating those that output o^* , these are few and μ^* can be selected with constant $O(1)$.

bits. The minimal program which runs the same computation has upper bounded complexity :

$$l(r^*) = K(\mu^* \mid o, K(o \mid a), a) \leq l(r'b'i^-j) = O(1) + l(b'i^-j) \quad (68)$$

With this, we have all the information which determines μ and we can bound from above its Kolmogorov complexity by considering a program z such that

$$U(z'i^+r^*j, o, a, K(o \mid a)) = i^+ \underbrace{T_r(b'T_{i^-}(j, o, a), K(o \mid a))}_{\mu^*} j = \mu$$

We can say z has constant length $l(z') = O(1)$ or $l(z') = O(\log m)$ since it just "routes" all the self-delimited information on the input tape and concatenates the output of T_r with i^+ and j . Finally, this means

$$\begin{aligned} l(z^*) &= \\ K(\mu \mid o, K(o \mid a), a) &\leq l(z') + l(r^*) + l(i^+j) \\ &\stackrel{(a)}{\leq} O(1) + l(b') + l(i^-j) + l(i^+j) \\ &\stackrel{(b)}{=} O(1) + l(l(i^-j) + l(a))' + 2l(i^-j) \\ &\stackrel{\pm}{=} l(l(i^-j) + l(a))' + 2l(i^-j) = c_\mu \end{aligned} \quad (69)$$

In (a) we applied the upper bound on $l(r^*)$ (Eq. 68) and in (b) the one on $l(b')$ (Eq. 67). Finally, we can obtain the following relationships.

$$c_\mu < K(\mu^*) \leq K(\mu) = K(\mu^*, i^+, j) \quad (70)$$

The second inequality holds because of [Hut24, Thm. 2.7.15]: μ contains μ^* and extra information. To demonstrate the first inequality, we can safely assume that the length of i^+ is small and constant (or $O(\log m)$ at most), while $K(\mu^*)$ can be arbitrarily larger than $2l(j)$. $\square$

This is further facilitated by the naturalistic assumption $l(o) \gg l(a)$ (As. I), which gives an exponentially large amount of c-incompressible [Li19, Def. 2.2.1] observation sequences:

$$K(\mu^*) \stackrel{\pm}{\geq} K(o_{\mu^*}) \geq l(o_{\mu^*}) - c \quad (71)$$

each with few unique minimal programs μ^* .

The construction adopted in the lemma above might seem quite specific. In fact, we might have a larger proportion of environments which do not have an action-independent incompressible component (See Appendix G). Nonetheless, the following proposition demonstrates that any environment could be represented with the same construction.

Proposition 8.3. *Every environment can be described/decomposed as in Lemma 8.2 with an increase in complexity bounded by $O(\log m) + O(\log |\mathcal{O}|)$.*

$$\forall \mu \in \mathcal{M}, \exists \boldsymbol{\mu} := \langle i^+, \mu^*, j \rangle :$$

$$\forall a : \mu(a) = \boldsymbol{\mu}(a) \quad \text{and} : K(\mu) + O(\log m) + O(\log |\mathcal{O}|) \stackrel{\pm}{\leq} K(\boldsymbol{\mu})$$

Proof. This can be trivially obtained by setting μ^* such that $\forall t : o_{t, \mu^*} = |\mathcal{O}|$ and $T_j = \mu$. With this :

$$\forall a, t : o_{t, \boldsymbol{\mu}} = |\mathcal{O}| + o_{t, j} \pmod{|\mathcal{O}|} = o_{t, j} = o_{t, \mu}$$

Then $K(\mu^*) \stackrel{\pm}{\leq} O(\log m) + O(\log |\mathcal{O}|)$ (it is a program which prints $|\mathcal{O}|$, or $0 \in \mathcal{O}$ if it exist, for m time steps). Noting that $K(i^+) \stackrel{\pm}{\leq} O(1)$ and $K(j) = K(\mu)$:

$$K(\boldsymbol{\mu}) = K(i^+, \mu^*, j) \stackrel{\pm}{\leq} K(i^+) + K(\mu^*) + K(j) \quad (72)$$

$$\stackrel{\pm}{\leq} O(\log m) + O(\log |\mathcal{O}|) + K(\mu) \quad (73)$$

$\square$

8.3 Interaction Between The Parameters of The Cybernetic Model

In this section we specify further how the parameters of our model (Def. 5.4) interact with each other and how they influence cost and ultimately, the behavior of the optimal agent.

Corollary 8.4 (Surprise is proportional to the complexity of the environment). *The surprise/IG, $S_{\mu \parallel \xi}^\pi$, given a fixed c_μ , is proportional to $K(\mu)$, until it reaches its upper bound ($S_{\mu \parallel \xi}^\pi \leq l(o)$)*

$$S_{\mu \parallel \xi}^{\pi, m} \propto K(\mu)$$

Proof. Altogether we have

$$\begin{aligned} K(\mu) - l(a) - 2 \log l(a) - c_\mu &\stackrel{(a)}{\leq} K(\mu \mid a) - c_\mu \\ &\stackrel{\pm}{=} K(\boldsymbol{\mu}') \stackrel{\pm}{=} S_{\mu \parallel \xi}^{\pi, m} \stackrel{(c)}{\leq} S_{\mu \parallel \xi}^\pi \stackrel{(d)}{\leq} S_{\mu \parallel \xi}^\pi \\ &\leq K(\mu) \end{aligned}$$

This is given by the fact that the bound in Thm. 8.1(a) scales with $K(\mu)$, while a $K(\mu)$ -dependent equality is given by Conj. 8.1(b) and Cor. 7.7(c). (d) follows by definition of optimal policy. Importantly, the inequality (a) holds by Thm. 8.1 even without tightness given by Conj. 8.1(b). $\square$

Since $c_\mu \stackrel{*}{=} l(i^+j)$ is fixed, the complexity of $K(\mu)$ can be increased only through $K(\mu^*)$.

Corollary 8.5 (Surprise is inversely proportional to actions dimensionality). *The surprise/IG $S_{\mu \parallel \xi}^\pi$ is inversely proportional to the dimensionality of the action space $\mathcal{A}^m$, until it reaches its lower bound ($S_{\mu \parallel \xi}^\pi \geq 0$)*

Proof. We can proceed with a reasoning analogous to Cor. 8.4. Moreover, $\min_a K(o \mid a)$ trivially decreases when the complexity is conditioned on a longer string. In fact, given two different action sequences A and a with different dimensionality such that $l(A) > l(a)$, if we reduce both actions sequences to a binary strings such that $\langle A \rangle = \langle a^* \rangle e$, where $a^* = \arg \min_a K(o \mid a)$ and e is a binary string with $l(e) = l(A) - l(a)$, we have:

$$\min_A K(o \mid A) \leq \min_e K(o \mid a^*e) \stackrel{(a)}{\leq} \min_a K(o \mid a) \quad (74)$$

Theorem 8.6. *The number of c -incompressible environments scales with $l(o) - l(a) = m \log \frac{|O|}{|A|}$*

Proof. In order to demonstrate this we could use Conj. G.1. Otherwise, we can write the following series of inequalities based on Eq. 66.

$$\forall a \quad K(o_{\mu^*} | a) \leq K(o | a) + l(i^-) + l(j)$$

$$\boxed{K(o_{\mu^*}) - l(a) - 2 \log l(a) - l(i^- j)} \leq K(o | a) \quad (75)$$

The boxed formula is a lower bound similar to the one presented in Thm. 8.1. We want the lower bound to be greater than $l(o) - c$ for the given observation sequence to be c -incompressible $\forall a$:

$$l(o) - c \leq K(o_{\mu^*}) - l(a) - 2 \log l(a) - l(i^- j) \leq K(o | a)$$

We can rewrite the above as follows.

$$l(o) + c' = l(o) - c + l(a) + 2 \log l(a) + l(i^- j) \leq K(o_{\mu^*}) \quad (76)$$

Then, we can count the number of sequences o_{μ^*} which respect this inequality. o_{μ^*} can be any sequence of length $l(o)$ for a total of $2^{l(o)}$ for which we can find a unique minimal program μ^* .

The number of programs of length less than $l(o) + c'$ is

$$\sum_{i=0}^{l(o)+c'-1} 2^i = 2^{l(o)+c'} - 1.$$

Therefore, there are at least :

$$2^{l(o)} - 2^{l(o)+c'} = 2^{l(o)} - 2^{l(o)-c+l(a)+2 \log l(a)+l(i^- j)}$$

strings, that have no program of length less than $l(o) + c'$ and an equal number of minimal programs μ^* for each such strings with $l(o) + 2 \log l(o) \geq K(o_{\mu^*}) = l(\mu^*) \pm K(\mu^*)$. This already demonstrates our proposition. Moreover, since j could be any valid program of length $l(j)$, each unique μ^* can be paired any j , without invalidating the above. We specify the number of valid programs with $|j|$. As a consequence, there are at least $|j|(2^{l(o)} - 2^{l(o)-c+l(a)+2 \log l(a)+l(i^- j)})$ environments $\mu = \langle i^+, \mu^*, j \rangle$ with $l(\mu) \leq l(o) + 2 \log l(o) + l(j) + l(i^+)$ such that $K(\mu') \geq l(o) - c$. $\square$

Finally, the following Corollary connects the lower bound presented in Thm. 8.1 to the relationships between information gain and surprise highlighted in Sec. D.

Corollary 8.7 (Quantifying the information gain). *Under the assumptions of Thm. 8.1, information gain and surprise are bounded from below:*

$$\begin{aligned} K(\mu) - l(a) - 2 \log l(a) - c_\mu &\leq_{(a)} K(\mu') \stackrel{\pm}{(b)} S_{\mu||\xi}^{\pi_{\mu||\xi}, m} =_{(c)} \mathbb{E}_{\mu}^{\pi_{\mu||\xi}}[IG_\mu] \\ &\leq_{(d)} \mathbb{E}_{\mu}^{\pi_{\mu||\xi}}[IG_\mu] \leq_{(e)} \frac{1}{w_\mu} \mathbb{E}_{\xi}^{\pi_{\mu||\xi}}[IG] \\ &\leq_{(f)} \frac{1}{w_\mu} \mathbb{E}_{\xi}^{\pi}[IG] \end{aligned}$$

²⁰(a) is a precise inequality if actions are on a separate input tape, otherwise we have to require $\langle A \rangle = \langle a^* \rangle' e$ and additive inequality so that e can be discarded if $K(o | a)$ is defined as:

$$K(o | A) = \min_p \{l(p) : U(A'p) = o\}$$

$\square$ *Proof.* (a) follows from Thm. 8.1, (b) from Thm. 7.7, (c) from Eq. 24, (d) by definition of optimal policy and (e) from Eq. 23: $\mathbb{E}_{\xi}^{\pi}[IG] = \sum_{\nu} w_{\nu} E_{\nu}^{\pi}[IG_{\nu}] \geq w_{\mu} E_{\mu}^{\pi}[IG_{\mu}]$. (f) follows by definition of optimal policy too. b,c,d,e, and f hold when conditioned on a history too. While for (a) this has yet to be verified. $\square$

9 The Agnostic Agent

9.1 Solomonoff's prior's Influence on Exploration and the AIXI agent

Section 8 outlined sufficient assumptions regarding the true environment that resolve the dark room problem. We did so by considering the μ -optimal agent which, knowing μ , explores less and experiences less surprise than the agnostic agent, thus providing a lower bound on the exploration of the latter.

In this section, we attempt to say something more regarding the behavior of the agnostic agent SMA (Def. 5.3), acting with policy $\pi_{\xi||\xi}$.

The future surprise, given the history generated by μ , has expected value at time t :

$$\mathbb{E}_{\mu}^{\pi_{\xi||\xi}} \left[S_{\xi||\xi}^{\pi_{\xi||\xi}}(h_{<t}) \right] \quad (77)$$

Since the past history is sampled from μ , $w_{\mu}(h_{<t})$ is monotone non-decreasing in time. Furthermore, there cannot be an environment μ'' with higher complexity and larger weight, since at any time t the following holds:

$$(K(\mu') < K(\mu) < K(\mu'')) \implies (w_{\mu'}(h_{<t}) > w_{\mu}(h_{<t}) > w_{\mu''}(h_{<t})) \quad (78)$$

In fact, in the deterministic setting, with each new observation any environment can be either discarded or kept, while weight are re-normalized with the sum of the prior of the remaining environments: $w_{\nu}(h) = \frac{w_{\nu}}{\mathbf{W}_{\mu}^0(h)}$. Therefore, the ordering given by Solomonoff prior is conserved in the posterior.

Now, consider Equation 22:

$$S_{\xi||\xi}^{\pi_{\xi||\xi}} = \sum_{\nu \in \mathcal{M}} w_{\nu} KL(P_{\nu}^{\pi_{\xi||\xi}} \| P_{\xi}^{\pi_{\xi||\xi}}) \quad (79)$$

The upper bound $S_{\nu||\xi}^{\pi} = KL(P_{\nu}^{\pi} \| P_{\xi}^{\pi}) \leq K(\nu)$ (Cor. 6.4) entails that the maximum contribution of each environment to the sum is ordered by their complexity.

$$K(\mu') < K(\mu) < K(\mu'')$$

$$\stackrel{(a)}{\implies} w_{\mu'} K(\mu') > w_{\mu} K(\mu) > w_{\mu''} K(\mu'')$$

$$\stackrel{(b)}{\implies} w_{\mu'}(h) K(\mu') > w_{\mu}(h) K(\mu) > w_{\mu''}(h) K(\mu'')$$

Implication (a) holds because $2^{-x}x > 2^{-y}y$ when $\frac{1}{\ln 2} \leq x < y^{21}$ and (b) holds because, as mentioned, the posterior is $\frac{w_{\nu}}{\mathbf{W}_{\mu}^0(h)}$, with a constant denominator for every environment ν .

This shows that, even if the maximum value for surprise/IG is smaller for simpler environments, its contribution to the expectation is exponentially greater than more complex environments. In fact, each bit of surprise in μ' weights $\frac{w_{\mu'}}{w_{\mu''}} = 2^{K(\mu'') - K(\mu')}$ times more than a bit of surprise in μ'' .

²¹In fact, the derivative $\frac{d(x \cdot 2^{-x})}{dx} = 2^{-x}(1 - x \ln 2)$ is ≤ 0 for $\frac{1}{\ln 2} \leq x$.

Example 9.1. If two policies π' and π'' differ only in their performance within these two environments, where π' performs better in μ' and π'' performs better in μ'' , such that:

$$d_{\mu'} = S_{\mu'|\xi}^{\pi'} - S_{\mu'|\xi}^{\pi''} > 0$$

and

$$d_{\mu''} = S_{\mu''|\xi}^{\pi'} - S_{\mu''|\xi}^{\pi''} > 0$$

Then, in order to prefer π'' , $\mathbf{sma}(\xi|\xi)$ would require that $d_{\mu''} \geq 2^{K(\mu'')-K(\mu')} d_{\mu'}$.

For instance, if we had $d_{\mu'} = 1$, $K(\mu'') = 8$ and $K(\mu') = 4$, in order to prefer π'' we would need $d_{\mu''} \geq 2^4$, which is not possible since:

$$d_{\mu''} \leq S_{\mu''|\xi}^{\pi'} \leq K(\mu'') = 8 = 2^3$$

In general, SMA can never prefer π'' if $K(\mu'') \leq 2^{K(\mu'')-K(\mu')} d_{\mu'}$.

Altogether, this suggests $\mathbf{sma}(\xi|\xi)$ will find a simple niche μ' similarly to $\mathbf{sma}(\mu|\xi)$. On the other hand, Appendix I reports an example which counter-argues this intuition. In fact, this example suggests how the agnostic agent, not knowing the true environment μ , can show even more interesting behavior in certain worlds, since its path to equilibrium is complicated by the fact that in its initial stage of life the agent will unwittingly discard potential niches.

Moreover, the following conjecture could help us pin down the fact that Solomonoff-based agents, having an infinite model class, expect unbounded surprise/knowledge over an infinite horizon, similarly to the KSA agent.

Conjecture 9.1.

$$\lim_{m \rightarrow \infty} S_{\xi|\xi}^{\pi, m} = \infty$$

Proof. This can be demonstrated similarly to [Ors13, Proposition 13]. We only need to make sure it holds for bigger action spaces too. $\square$

9.2 Asymptotic Optimality

While in the above sections we focused mainly on a “single episode” of the agent’s experience up to the horizon m with no history, in the following we elaborate on the long-term behavior of SMA, conditioned on the history. In AIXI territory, a commonly discussed property is *asymptotic optimality*: whether the agnostic agent, here $\mathbf{sma}(\xi|\xi)$, approaches the μ -optimal policy in the limit, effectively learning it. While there are two theorems in [Hut24] against strong (Thm. 8.3.1) and weak (Thm. 8.3.2) asymptotic optimality, these can be invalidated with asymptotically optimal stochastic agents such as the already mentioned InQ (Thm. 9.4.2) and BayesExp (Thm. 9.4.1). Furthermore, [Ors14] proves weak and strong optimality for the KSA agent in the deterministic setting with no modification to the regular *deterministic* AIXI expectimax policy using negative likelihood as reward (maximizing surprise). This is why we believed something similar were achievable. We adapt the definition of optimality in mean [Hut24, Def. 8.1.6] for the difference to be positive (we are doing minimization instead of maximization so the expected cost given by $\mathbf{sma}(\xi|\xi)$ will be greater) than the one given by $\mathbf{sma}(\mu|\xi)$.

Theorem 9.1. $\mathbf{sma}(\xi|\xi)$ is asymptotically optimal in mean in deterministic environments:

$$\lim_{t \rightarrow \infty} \Gamma_t^{-1} \mathbb{E}_\mu^{\pi_{\xi|\xi}} \left[S_{\mu|\xi}^{\pi_{\xi|\xi}}(h_{<t}) - S_{\mu|\xi}^{\pi_{\mu|\xi}}(h_{<t}) \right] = 0 \quad (80)$$

Proof. When μ is deterministic, it holds from Eq. 25 that expected surprise is equal to step-wise discounted KL divergence.

$$S_{\mu|\xi}^{\pi_{\xi|\xi}}(h) = KL_h^\gamma(P_\mu^{\pi_{\xi|\xi}} \| P_\xi^{\pi_{\xi|\xi}}) \quad (81)$$

Thanks to [Ors14, Thm. 5] we know that, based on the Solomonoff bound[Hut24, Thm. 3.2.5], the above divergence converges to zero on-policy:

$$\lim_{n \rightarrow \infty} \Gamma_t^{-1} \mathbb{E}_\mu^{\pi_{\xi|\xi}} KL_{h_{<t}}^\gamma(P_\mu^{\pi_{\xi|\xi}} \| P_\xi^{\pi_{\xi|\xi}}) = 0 \quad (82)$$

Now, we can demonstrate our theorem combining the two equations above together with the fact that $S_{\mu|\xi}^{\pi_{\xi|\xi}}(h_{<t}) \geq S_{\mu|\xi}^{\pi_{\mu|\xi}}(h_{<t})$ by definition of optimal policy, which entails a positive difference (a).

(Eq.80) =

$$\underbrace{\lim_{t \rightarrow \infty} \Gamma_t^{-1} \mathbb{E}_\mu^{\pi_{\xi|\xi}} \left[S_{\mu|\xi}^{\pi_{\xi|\xi}}(h_{<t}) \right] - \Gamma_t^{-1} \mathbb{E}_\mu^{\pi_{\xi|\xi}} \left[S_{\mu|\xi}^{\pi_{\mu|\xi}}(h_{<t}) \right]}_{=0 \text{ (Eqs. 81 \& 82)}} \stackrel{(a)}{\geq} 0$$

$\square$

9.3 Regret Minimization

“A data-efficient agent must select actions that suitably balance between exploration and exploitation. An optimal balance can be achieved by designing the agent to minimize regret” [Lu23] While asymptotic optimality tells us something about the long term behavior of the agent, it gives no guarantees in terms of the rate of convergence to the optimal policy. This section fills in this gap by analyzing the regret of the agnostic agent.

There are two different ways in which we can think of regret in this setting. The first is the difference in total prediction error in the first m steps with respect to the prediction error given by μ itself, which is ultimately the best predictor since observations are sampled from the latter.

Then we would like to find the policy π which minimizes the KL divergence:

$$\begin{aligned} KL^{\pi, m}(\mu|\xi) &= \sum_{h \in \mathcal{H}^m} P_\mu^\pi(h) \left[\log \frac{P_\mu^\pi(h)}{P_\xi^\pi(h)} \right] \\ &\stackrel{(a)}{=} \sum_{h \in \mathcal{H}^m} P_\mu^\pi(h) \left[\log \frac{\mu(o|h)}{\xi(o|h)} \right] \\ &= S_{\mu|\xi}^{\pi, m} - S_{\mu|\mu}^{\pi, m} \\ &\stackrel{(b)}{=} S_{\mu|\xi}^{\pi, m} \stackrel{(c)}{=} \mathbb{E}_\mu^\pi [IG_\mu] \end{aligned}$$

With (a) we canceled out the terms $\pi(a_k | h_{<k})$ of the chronological distributions in the fraction. (b) holds because $S_{\mu|\mu}^{\pi, m}$ is zero in the deterministic setting, while (c) holds because of Eq.24. $\mathbf{sma}(\mu|\xi)$ minimizes exactly this KL in the deterministic setting, while $\mathbf{sma}(\xi|\xi)$ does so in a balanced Pareto optimal way (Thm. 4.1). On the other hand, we are also interested in the regret given by the policy, rather than the one stemming from the predictor, for which we already have nice convergence result under any policy[Hut24, Thm. 3.2.5, Generalized Solomonoff bound] For this reason, we adapt for minimization the definition of regret ([Hut24, Def. 8.1.9]) used within UAI and compare the optimal policy with the agnostic policy over the predictor we have at hand, ξ .

$$\begin{aligned} \text{Regret}_m(\pi_{\xi||\xi}, \mu) &:= S_{\mu||\xi}^{\pi_{\xi||\xi}, m} - \inf_{\pi} S_{\mu||\xi}^{\pi, m} \\ &= S_{\mu||\xi}^{\pi_{\xi||\xi}, m} - S_{\mu||\xi}^{\pi_{\mu||\xi}, m} \end{aligned} \quad (83)$$

This practically represents the difference in the complexity of the minimal niche found by $\mathbf{sma}(\xi||\xi)$ and $\mathbf{sma}(\mu||\xi)$ respectively.

Theorem 9.2. (Sublinear Regret) Assuming the true environment μ does not change with time, $\mathbf{sma}(\xi||\xi)$ has sublinear regret.

$$\text{Regret}_m(\pi_{\xi||\xi}, \mu) = o(m)$$

Proof.

$$\begin{aligned} \text{Regret}_m(\pi_{\xi||\xi}, \mu) &= S_{\mu||\xi}^{\pi_{\xi||\xi}, m} - S_{\mu||\xi}^{\pi_{\mu||\xi}, m} \\ &\stackrel{(a)}{=} K(\mu'_{a_{\xi||\xi}}) - K(\mu') \\ &\leq K(\mu) \end{aligned} \quad \begin{matrix} (b) \end{matrix}$$

(a) because of Thm. 7.6 and Lem. 7.4. (b) Since $K(\mu') \leq K(\mu'_{a_{\xi||\xi}}) \leq K(\mu)$. Now, if μ does not change with time, $K(\mu)$ has a constant value and does not increase with $m \rightarrow \infty$.

We might have to specify the following lower bound since μ is represented by a program which halts which prints a program of length $l(o) \in \mathbb{N}^+$ which depends on m .

$$K(l(o) | a) \leq K(\mu) + O(1)$$

Nonetheless:

$$\begin{aligned} K(l(o) | a) &\stackrel{(a)}{\leq} K(l(o)) \stackrel{(b)}{\leq} \log l(o) + 2 \log \log l(o) \\ &\leq 2 \log(m) + 2 \log \log |\mathcal{O}| \end{aligned}$$

For the last equality we used $\log(x) + 2 \log \log(x) \leq 2 \log(x)$ and $l(o) = m \log |\mathcal{O}|$

□

Thm. 9.2 suggests that, under SMA's cost (or negative reward), all environments with finite complexity are recoverable and that the $\pi_{\xi||\xi}$ policy is asymptotically optimal in mean (Thm. 9.1), since theorem [Hut24, p. 8.1.13] demonstrates the implication in the opposite direction. In fact, while normally we want to avoid "traps", here they represent the desired outcome, since they provide limited surprise once the agent has entered them.

9.4 Probability of Dying

What originally motivated surprise minimization is the biological prerogative of survival of embedded agents. Therefore, we deem it important to reflect whether it is effectively respected by SMA using few of the already developed theorems on the subject for AIXI. [Mar16] defines the probability of death at time t given by semi-measure ρ with the measure-loss²² as follows:

$$L_{\rho}(h_{<t}a_t) := 1 - \sum_{o_t} \rho(o_t | h_{<t}a_t)$$

²²This quantifies the total missing probability mass that would allow the sum of probabilities of all outcomes to sum up to 1

This represents the posterior probability mass of the non-halting environments $\nu \in \mathcal{M}$. We can consider the non-halting of the unknown μ as the death of the agent since it can't receive any new observation [Mar16]. This can be interpreted as the dissipation of the Markov blanket (Fig. 1) which maintains conditional independence between internal and external states and allows for the observation-action cycle described in As. 5.3.

If we take $\xi = \rho$ and minimize the semi-measure loss, we obtain the following.

$$\arg \min_{a_t} 1 - \underbrace{\sum_{o_t} \xi(o_t | h_{<t}a_t)}_{L_{\xi}} = \arg \min_{a_t} - \log \sum_{o_t} \xi(o_t | h_{<t}a_t)$$

Indeed, the logarithm does not affect the minimum due to its monotonicity. We can now reconstruct the expected cost of the $\mathbf{sma}(\xi||\xi)$ agent, by moving the expectation outside of the logarithm. Given $Z_{a_t} := \sum_{o_t} \xi(o_t | h_{<t}a_t)$, consider the following equality.

$$\begin{aligned} & - \log Z_{a_t} + H \left[\frac{\xi(o_t | h_{<t}a_t)}{Z_{a_t}} \right] \\ &= \sum_{o_t} \frac{\xi(o_t | h_{<t}a_t)}{Z_{a_t}} \left[- \log \left(\frac{\xi(o_t | h_{<t}a_t)}{Z_{a_t}} \right) \right] \end{aligned}$$

Then, by positivity of entropy and simplifying the fraction within the log term, we obtain the following inequality.

$$- \log \sum_{o_t} \xi(o_t | h_{<t}a_t) \leq \sum_{o_t} \frac{\xi(o_t | h_{<t}a_t)}{Z_{a_t}} [- \log \xi(o_t | h_{<t}a_t)]$$

Finally, applying the minimum on both sides, we get

$$\begin{aligned} \min_{a_t} - \log \sum_{o_t} \xi(o_t | h_{<t}a_t) &\leq \min_{a_t} \sum_{o_t} \frac{\xi(o_t | h_{<t}a_t)}{Z_{a_t}} [- \log \xi(o_t | h_{<t}a_t)] \\ &\stackrel{(a)}{=} \min_{a_t} \frac{S_{\xi||\xi}^{\pi, 1}}{Z_{a_t}} \end{aligned}$$

(a) holds considering a deterministic policy $\pi(a_t | h_{<t}) = 1$, which picks the selected action. The reasoning above can be extended to multi-step minimization for long-term survival and it suggests that it might be best to normalize the distribution used for sampling expected observations (ρ in Def. 5.2), when we are dealing with a semi-measure. Nonetheless, the normalization factor would not significantly alter the results presented in earlier sections, and this potential adjustment is left for future work.



We could demonstrate that $\mathbf{sma}(\xi||\xi)$ is balanced Pareto optimal for probability of death minimization.

10 Discussion

10.1 Summing Up

While we initially hypothesized that a "knowledge-seeking" exploratory behavior would organically emerge from the objective of surprise minimization, Cor. 7.7 proved that SMA is *actively* seeking compressible future observation sequences. In other words, it acts on the environment to induce an observation

sequence described by a short program, given its reference Turing machine U .

Nonetheless, with Thm. 8.1 we provided a set of naturalistic environmental properties, under which SMA could not immediately converge to a low complexity niche, and we outlined a construction which respects these properties (Lem. 8.2). The key characteristics of this type of environments are that the agent has limited control over its environment, which fits with the big world hypothesis [Jav24], and that observations are much more “expressive” as compared to actions.

As stated, the goal of this work was not to prescribe the objective function that an optimal agent should have, but rather characterize the interaction between parameters (e.g. prior, observation, actions and environment space: Def. 5.4) of the cybernetic model for an embedded agent which performs perfect Bayesian inference. We have shown in Sec. 8.3 how these parameters influence the surprise and information gain experienced by the agent. For instance, on a shorter horizon SMA would settle on a simpler niche than the one it would need to find, by exploring, with a longer horizon (re-interpreting Thm. 8.6).

Finally, in Sec. 9 we demonstrated a set of generally desirable properties for any agnostic agent, namely asymptotic optimality, sublinear regret, and probability-of-death minimization.

Altogether, this suggests that the objective of surprise minimization is worth exploring further.

Moreover, the analysis presented here might be applicable to any embedded agent, whenever the reward function can be encoded in the model’s prior.

10.2 The Environment is The Teacher

With our analysis we have highlighted several important limits of embedded agents. Firstly, when the prior is far from the true environment, it will induce a sort of “delusion”, because once the agents reaches a local optimum, represented by a niche from which no reward path signals safe and beneficial exploration out of it, it will likely stay in it. Furthermore, there are combinatorially many such local minima if the modeled environment depends heavily on the action sequence. In fact, only in the passive setting, where the agent cannot act, the Solomonoff predictor provably converges to the true environment with a limited cumulative error [Hut24, Thm. 3.2.5]. In an agentic setting instead, only by trying all possible action sequences, as KSA does, the model will eventually learn μ off-policy (Thm. 7 [Ors13]).

This might be clearly infeasible in practical application and risky for embedded agents which can’t experience death multiple times in order to learn which actions lead to it. The abstract concept of death needs to be learned and modeled well before the world presents risky situations to the agent. In order for this to happen, observations need to be informative about the environment explaining, in great part, the dynamics of the local surroundings: In Thm. 8.1 and Lem. 8.2 the action-independent component of the environment ultimately determines the complexity of the observations and entailed niche.²³ We could argue that no general algorithm

²³A naive way for this to be achieved is by increasing the dimensionality of $\mathcal{O}$, which clearly induces higher computational costs. This is probably not necessary anyway as the information gained from them needs to be effectively useful and reflect the future of the agent, allowing the latter to adapt to a new niche if the present changes. Ultimately, the agent is heavily dependent on the niches it ends up in and how informative the observations are within and around it.

can perform optimally in any environment if the latter doesn’t organically and safely teach to the agent²⁴. In fact, the agent cannot know a priori how safe it is to explore in a specific environment unless it has a specific prior or inductive bias for it. And if it does already know the true environment, it means that little exploration is needed. Within the environment-mentor case, the reward function loses importance if the teacher knows how to exploit it in order to teach to the agent²⁵.

Altogether, agency seems to be deeply tied to the scale of the sensory-motor interface and the extent to which the environment favors learning, in addition to internal computational limitations and the efficiency of the agent.

10.3 Bounded Approximations

Solomonoff induction and expectimax (expectimin in this case) rely on unbounded computational resources (As. 5.2). In this section we discuss the computability of SMA, how it relates to real-world models, and potential ways to reduce its computational burden.

10.3.1 Computability of SMA

[Lei18, Cor. 36] has already demonstrated that entropy-seeking (Shannon-KSA) and knowledge-seeking (KSA) ε -optimal policies are limit computable, or Δ_2^0 -computable²⁶, for a finite horizon.

Likewise, we can say that the ε -optimal policy for the $\mathbf{sma}(\xi||\xi)$ agent is limit-computable. This is motivated by the fact that the expected cost $S_{\xi||\xi}^{\pi, m}$, is itself limit-computable, since it is composed of limit-computable functions of the form $\xi(\cdot|h_{<k}a_k)$, combined through computable operations ($+$, $-$, $\cdot$ and $\log$) [Lei18, Lem. 2] over a finite number of possible future interaction sequences $h_{t:m}$ (Eq. 18). In fact, we can approximate with ε precision the optimal cost $S_{\xi||\xi}^{\pi, m}$ by considering the finite number of all deterministic policies. By limit-computing their expected cost, the lowest so far will eventually be within ε from the optimal cost²⁷.

Nevertheless, the universal SMA agent is practically unfeasible. To approximate it in a computable manner we could replace the universal distribution ξ with a mixture over a bounded class of models C , which can be used to probabilistically approximate [Blo14] the complexity of a sequence o generated by a model $\nu \in C$ within the class: $K(o | a) \stackrel{\pm}{=} S_{\nu||\xi}^{\pi}$ (Thm. 7.6). At the same time, using a modern neural network instead of a Bayesian mixture might represent a more practical solution. In fact, these have been shown to implicitly respect Occam’s razor and the MDL principle [Bli18].

10.3.2 Modern AI Models

We can draw a few connections with state-of-the-art attention-based models. These, like SMA, rely on a cross-entropy loss on

²⁴See [Coh21] for an AIXI variant, Mentee, where the agent’s exploration is guided by a mentor.

²⁵It would come as no surprise if evolution chose prediction loss as the negative reward for biological organisms. In fact, their habitat on earth is usually chaotic and dense with such signals.

²⁶A function $f(x)$ is limit-computable if there is a recursive (computable) function, $\phi(x, k)$, such that $\lim_{k \rightarrow \infty} \phi(x, k) = f(x)$. To learn about computability levels and arithmetic hierarchies the reader may refer to [Hut24, Sec. 2.6]

²⁷In fact, as already said, we are guaranteed to have a deterministic policy in the set of optimal policies. This might become clearer considering the expected cost in the recursive form as in Eq. 89.

next-token (here observation) prediction. Furthermore, one of the many reasons why pre-training is so effective might be that in this phase the model cannot directly act on its environment, while being passively exposed to a large quantity of information (with high Kolmogorov complexity). Moreover, within this process the action space is empty, while the observation space gets vaster as the training batch gets larger. Altogether, this implies an increasing lower bound on the information gain, in line with Thm. 8.1 and Cor. 8.4.

10.3.3 Alternatives to Expectimax

While in Sec. 10.3.1 we proposed approaches to make the prediction and computation of the cost function feasible, we didn't address how we might relieve the computational burden for planning. In fact, currently we are simply considering all possible futures and actions in order to find the optimal policy. This is clearly exponentially intractable as we have to compute the expected cost for $|\mathcal{H}|^m$ possible interaction sequences. In order to avoid this, we could take inspiration from the "action as inference" idea underlying most FEP formulations, which enables (variational) inference on actions too [Vri25], as afforded by other frameworks [Lev18] in optimal control theory. To this end, while staying in the realm of UAI, we could research a surprise-minimizing version of self-AIXI [Cat23]. This agent adopts an additional Bayesian mixture ζ to self-predict its own policy. A surprise-minimizing self-AIXI might have a different behavior especially if we properly integrated ζ in Eq. S[h] to effectively encode action uncertainty within the cost function; E.g:

$$\begin{aligned} S_{\varrho}^{\pi}(h_t \mid h_{<t}) &= -\log P_{\varrho}^{\zeta}(h_t \mid h_{<t}) \\ &= -\log[\zeta(a_t \mid h_{<t})\varrho(o_t \mid h_{<t}a_t)] \end{aligned}$$

10.3.4 Bounded Memory and Latent state

Currently our agent has infinite memory. This is not necessarily a strong assumption, since we can easily imagine an artificial agent having enough memory to store the full interaction sequence spanning over its lifetime. Nonetheless, this indirectly influences the computational costs as the more the agent perceives, the more complex its model of the world will have to become: $w_{\nu}(h)$ increases monotonically for complex ν s. In order to overcome this, we might decrease the computational burden given by an extensive model class, such as $\mathcal{M}_{det}$ in this case. In fact, as anticipated in Sec. 4, instead of trying to model the environment in its entirety, the latter could be described as a latent state holding the information relevant to the agent. In fact, the real world might be too complex for any biological or artificial agent to model in its entirety, and it is much more efficient to give greater attention to the local surroundings instead. This idea is already captured fairly well by the concept of niche (Def. 7.1). With this in mind, we could modify the variational formulation proposed in Sec. 4.2 to accommodate for an effective latent state η which changes with time. More precisely, η can represent the minimal niche μ' which can effectively model the future of the agent up to horizon m (Eq. 36). The posterior beliefs of the agent, over the minimal niche, might be written as $q(\eta = \mu' \mid o, a, w) = \sum_{\nu} w_{\nu}(h) [[\nu' = \mu']]$. Where $[[\nu' = \mu']]$ has a value of 1 if the minimal niches in environment ν and μ' , respectively, are equal. As mentioned, we do not necessarily care for the niche model to explain the past of the agent too,

the formulation above only serves as an example, and a more precise analytical formulation can be devised.

10.4 Alternative Prior Or Intrinsic motivation

It might be interesting to combine the surprise-minimizing behavior of SMA with particular preferences on certain observations or histories. These can be expressed as regular rewards [Bow23] or encoded within the agent's model as a bias towards preferred percepts. The latter is often done in active inference literature as it allows to integrate arbitrary reward signals without altering the main motivation of realizing the model's predictions [Mil20]. Finally, it would be insightful to clarify how the information gain maximization term, which can emerge from the regular free energy formulations in dynamical systems [Fri231; Fri232], relates to the definition of IG in Eq. 3. The latter is instead minimized here as the cost the function was derived from a variational upper bound (negative ELBO) on surprise (Thm. 4.1). In fact, in order for the IG term to appear with a negative sign, free energy has to be formulated as a lower bound on surprise. This distinction has been already highlighted in [Mil21] where a novel formulation was proposed to integrate the epistemic motivation while maintaining a theoretical grounding as a continuation of variational inference in the future. It is referred to as Free Energy of the Expected Future and corresponds to the minimization of the KL divergence between a biased generative model, encoding the preference of the agent, and a veridical model. This objective function might be interesting to explore in the context of UAI as a continuation of the research carried out here. In fact, SMA represents the baseline case where the agent has no preference towards particular observations as long as they are realized (the agent has survived). Nonetheless, as demonstrated here, the agent displays exploratory behavior even with the simplest goal of surprise minimization. This entails that it is not necessary for a knowledge seeking term to appear explicitly in the objective function for it to naturally arise.

11 Conclusion

Hopefully, the discussion above has already suggested inspiring directions for future research. Furthermore, the reader might also attempt to resolve the conjectures hereby presented.

12 Acknowledgments

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13 Use of Generative AI

Being at the intersection of FEP and UAI, the topic of this thesis is quite out of distribution. Any attempt to use AI to generate novel proofs has proven counterproductive, and no AI-generated content appears in the final report. Nonetheless, GPT 5.0 was used while I was still learning about these two

²⁸uaiasi.com

fields to help me understand some concepts. It was also used to find useful resources or relevant chapters from longer books, such as [Li19]. This model also helped me in noticing some mistakes or discouraging some directions in earlier proof attempts. Finally, Claude Opus 4.7 was used for a final check on syntactic, grammatical or obvious conceptual mistakes.

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A Deriving The Cost Function from the Free Energy Principle

Here we continue the discourse of Section 4.2 in order to demonstrate Thm. 4.1.

Theorem 4.1 (Expected Surprise Minimization). *Given assumptions 4.1, 4.4 and 4.5, minimizing the variational bound on expected surprise over future trajectories o, a, w , governed by true dynamics $p(o, a, w, \eta)$, is equivalent to the minimization of expected surprise with respect to $q(o, a, w)$. Given this objective, the universal mixture ξ is balanced Pareto optimal with respect to Solomonoff’s prior.*

$$\arg \min_{a, w} \mathbb{E}_{o \sim p(\cdot|a, w)} [F(o, a, w)] = \min_{\pi} \mathbb{E}_{P_{\pi}^{\pi}(h)} [-\log P_q^{\pi}(h)] \quad (i)$$

$$\begin{aligned} & \min_{q, \pi} \mathbb{E}_{\mu \sim \xi(\cdot)} \mathbb{E}_{h \sim P_{\mu}^{\pi}(h)} [-\log P_q^{\pi}(h)] \\ & = \min_{\pi} \mathbb{E}_{\mu \sim \xi(\cdot)} \mathbb{E}_{h \sim P_{\mu}^{\pi}(h)} [-\log P_{\xi}^{\pi}(h)] \end{aligned} \quad (ii)$$

Proof.

Motivating (i)

First, consider Eq. 6. If the agent knew future observations beforehand, it could select the action sequence which realizes them. Since this is rarely the case, it can only minimize expected surprise minimization over state trajectories.

$$\mathbb{E}_{o \sim p} \left[\min_{a, w} -\log p(o, a, w) \right] \stackrel{(a)}{=} \min_{a, w} \mathbb{E}_{o \sim p} [-\log p(o, a, w)] \quad (84)$$

Where (a) holds because Jensen’s inequality turns into an equality when $p(o, a, w)$ is not stochastic. The above value is zero and any deterministic dynamics $p(a, w)$ which maintain the Markov blanket, thus producing some observations, would fulfill surprise minimization with respect to the true distribution. In fact, having the correct prior on the true environment, p already “knows” future observations.

Things become interesting when we write the minimization of $-\log p(o, a, w)$ from the perspective of the agent, as a variational upper bound given by its biased distribution q .

$$\mathbb{E}_{o \sim p} [-\log p(o, a, w)] \stackrel{(a)}{\leq} \mathbb{E}_{o \sim p(\cdot|a, w)} [F(o, \alpha)] \quad (85)$$

$$\stackrel{(b)}{=} \mathbb{E}_{o \sim p(\cdot|a, w)} \left[\mathbb{E}_{\eta \sim q(\cdot|o, a, w)} \log \left[\frac{q(\eta|o, a, w)}{p(\eta, o, a, w)} \right] \right] \quad (86)$$

$$= \mathbb{E}_{p(\cdot|a, w)q(\cdot|o, a, w)} \log \left[\frac{q(o, a, w | \eta)q(\eta)}{q(o, a, w)p(o, a, w | \eta)p(\eta)} \right] \quad (87)$$

$$= \underbrace{\mathbb{E}_{p(o|a, w)} [-\log q(o, a, w)]}_{\text{cross-entropy}} + \underbrace{KL[q(\nu)||p(\eta)]}_{\text{constant}} \quad (88)$$

(a) holds because F here represents the negative ELBO over $-\log p(o, a, w)$, this can also be referred to as free energy of the future (FEF)[Mil21], as it is slightly different from the usual expected free energy formulation (EFE), which we will come back to in the discussion section. In Eq. 87 we cancel out the generative models because, as already mentioned, their equality is afforded by As. 4.5.

The derivation above shows that the upper bound on true surprise given by the FEF (Eq. 85) simplifies to cross-entropy plus the KL divergence between the priors. Therefore, we can directly minimize this cross-entropy. Which is a well known upper bound on the entropy, or expected surprise, given by the true distribution.

Since we are working in sequential setting (As. 5.3) and we have uncertainty over observations, at each new time step t

the agent has access to a longer history h of observations and actions. Therefore, we can conditionally minimize the bound over new incoming observations.

$$\begin{aligned} \min_{a,w} \mathbb{E}_{p(o|a,w)} [-\log q(o, a, w)] &= \\ \min_{a_1, w_1} \mathbb{E}_{p(o_1|a_1, w_1)} &\left[-\log q(o_1, a_1, w_1) + \right. \\ \min_{a_2, w_2} \mathbb{E}_{p(o_2|h_{<2}a_2, w_2)} &\left[-\log q(o_2, a_2, w_2|h_{<2}) + \dots \right] \\ &= \min_{\pi} \mathbb{E}_{P_p^\pi(h)} [-\log P_q^\pi(h)] \end{aligned} \quad (89)$$

Where $P_q^\pi(h) = q(o, a, w)$ is the chronological distribution defined in Sec. 2.5 using environmental distribution $q(o_t | h_{<t}a_t, w_t) = q_{w_t}(o_t | h_{<t}a_t)$. We leave aside the term representing internal states w , which parametrize q , since they are deterministically given by the Bayesian posterior calculation. $P_p^\pi(h) = p(o | a, w)$ holds considering that π is the deterministic policy which takes (optimal) actions a , and o depends only on a under p .

Finally, (i) follows from Eq. 85, 87, 88 and 89.

Motivating (ii) Since the agent does not have access to the true prior $p(\eta)$, we cannot use the distribution $p(o | a, w)$ that gives the *true* expected surprise:

$$\mathbb{E}_{q(o|\alpha)} [-\log q(o, a, w)] \neq \mathbb{E}_{p(o|\alpha)} [-\log q(o, a, w)] \quad (90)$$

Nonetheless, we can use the universal mixture ξ (Sec. 2.6) as our q , which is balanced Pareto optimal with respect to its own prior.

In fact, for every policy π we can set a performance measure $\mathcal{F}$ as the negative log-likelihood given by a predictor $q(\cdot | h_{<t}a_t) \in \mathcal{M}$, with respect to the observations, which are deterministically generated by an environmental distribution $\mu(\cdot | h_{<t}a_t) = p(\cdot | h_{<t}a_t, \eta = \mu) = p(\cdot | h_{<t}a_t)$:

$$\mathcal{F}(\mu, q) = \mathbb{E}_{h \sim P_\mu^\pi(h)} [-\log P_q^\pi(h)] \quad (91)$$

from [Hut24, Th. 3.6.4] we have that the universal mixture ξ (Sec. 2.6) predictor ξ is balanced Pareto optimal with respect to its own prior, w_μ :

$$\forall \varrho : \sum_{\mu \in \mathcal{M}} w_\mu (\mathcal{F}(\mu, q) - \mathcal{F}(\mu, \xi)) \geq 0$$

From the above equation, we have:

$$\sum_{\mu \in \mathcal{M}} w_\mu \mathcal{F}(\mu, q) \geq \sum_{\mu \in \mathcal{M}} w_\mu \mathcal{F}(\mu, \xi)$$

Substituting $w_\mu = \xi(\mu)$ and the definition of $\mathcal{F}$ (Eq. 91), we obtain:

$$\begin{aligned} &\forall \pi, q \in \mathcal{M} \\ &\mathbb{E}_{\mu \sim \xi(\cdot)} \mathbb{E}_{h \sim P_\mu^\pi(h)} [-\log P_q^\pi(h)] \\ &\geq \mathbb{E}_{\mu \sim \xi(\cdot)} \mathbb{E}_{h \sim P_\mu^\pi(h)} [-\log P_\xi^\pi(h)] \end{aligned} \quad (92)$$

□

We could have motivated (ii) simply with

$$\mathbb{E}_{\mu \sim \xi(\eta)} \mathbb{E}_{h \sim P_\mu^\pi(h)} [-\log P_\xi^\pi(h)] = \mathbb{E}_{h \sim P_\xi^\pi(h)} [-\log P_\xi^\pi(h)] \quad (93)$$

And

$$\forall q : \mathbb{E}_{h \sim P_\xi^\pi(h)} [-\log P_\xi^\pi(h)] \leq \mathbb{E}_{h \sim P_q^\pi(h)} [-\log P_q^\pi(h)] \quad (94)$$

However, we believe the given proof provides a slightly more compelling story. Given Eq. 93, ii expresses the same quantity minimized by our agent (Eq. 19).

B Proofs of Simpler Bounds on Expected Cost

B.1 Lower bound on expected surprise

We have the lower bound $0 \leq S_{\rho||\varrho}, \gamma$ for any (semi)measures ρ, ϱ and discount $\gamma \geq 0$ since $\varrho(\cdot) \leq 1$ because the latter is a distribution. Finally, $-\log x \geq 0$ for $x \leq 1$; where equality $-\log x = 0$ holds for $x = 1$.

B.2 Upper bound on expected surprise

Lemma B.1.

$$\forall \pi, \varrho, \gamma : \Gamma_t < \infty \quad S_{\varrho||\varrho}^{\pi, \gamma} \leq \underbrace{\log |\mathcal{O}| \Gamma_t}_{\star} \text{ or } \underbrace{\log |\mathcal{H}| \Gamma_t}_{\diamond} \quad (95)$$

Where the upper bound is $\star$ if cost is set as $S[o]$ or $\diamond$ with $S[h]$.

Proof. We can write expected cost, from Eq. 13, as:

$$S_{\varrho||\varrho}^{\pi}(h_{<t}) = \sum_{k=t}^{\infty} \gamma_k \underbrace{\sum_{h_{t:k-1} \in \mathcal{H}^{k-t}} P_{\varrho}^{\pi}(h_{t:k-1} | h_{<t})}_{\diamond} \underbrace{\sum_{h \in \mathcal{H}} P_{\varrho}^{\pi}(h_k | h_{<k}) [S_{\varrho, k}]}_{\blacktriangle} \quad (96)$$

Now we have that $\diamond \leq 1$ and, if cost set as surprise on action-observation pair($S[h]$):

$$\begin{aligned} \blacktriangle &= \sum_{h \in \mathcal{H}} P_{\varrho}^{\pi}(h_k | h_{<k}) [-\log P_{\varrho}^{\pi}(h_k | h_{<k})] \\ &\stackrel{(a)}{\leq} \sum_{h \in \mathcal{H}} P_{\varrho}^{\pi}(h_k | h_{<k}) \left[-\log \frac{1}{|\mathcal{H}|} \right] \\ &\leq \log \mathcal{H} \end{aligned}$$

Where (a) holds by Gibb's inequality with the cross entropy of the uniform distribution.

Instead, with $S[o]$:

$$\begin{aligned} \blacktriangle &= \sum_{h \in \mathcal{H}} P_{\varrho}^{\pi}(h_k | h_{<k}) [-\log \varrho(o_k | h_{<k} a_k)] \\ &= \underbrace{\sum_{a_k \in \mathcal{A}} \pi(a_k | h_{<k})}_{\leq 1} \underbrace{\sum_{o_k \in \mathcal{O}} \varrho(o_k | h_{<k} a_k) [-\log \varrho(o_k | h_{<k} a_k)]}_{\substack{\leq \log \mathcal{O} \\ (b)}} \\ &\leq \log \mathcal{O} \end{aligned}$$

(b) using Gibb's inequality with the uniform distribution as above.

Altogether with Eq. 96 we have:

$$S_{\varrho||\varrho}^{\pi, \gamma} \leq \sum_{k=t}^{\infty} \gamma_k \log |\mathcal{H} \text{ or } \mathcal{O}| = \log |\mathcal{H} \text{ or } \mathcal{O}| \Gamma_t$$

□

C Telescoping Property of Surprise

Proof. Given $\gamma_k = [[k \leq m]]$, we provide the proof for the general case, with $\rho \neq \varrho$, and we refer to per step surprise $S_{\varrho,k} = -\log[\varrho(o_k | h_{<k}a_k)]$ OR $P_\rho^\pi(h_k | h_{<k})$ at time k which is agnostic of whether we use Eq. S[o] or S[h]. First, we can rewrite the cost function, leaving out the optional normalization factor Γ_t :

$$S_{\rho||\varrho}^{\pi,m}(h_{<t}) \stackrel{(a)}{=} \sum_{k=t}^m 1 \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\rho^\pi(h_{t:k} | h_{<t}) [S_{\varrho,k}] \quad (97)$$

$$\stackrel{(b)}{=} \sum_{k=t}^m 1 \sum_{h_{t:m} \in \mathcal{H}^m} P_\rho^\pi(h_{t:m} | h_{<t}) [S_{\varrho,k}] \quad (98)$$

$$\stackrel{(c)}{=} \sum_{h_{t:m} \in \mathcal{H}^m} P_\rho^\pi(h_{t:m} | h_{<t}) \left[\sum_{k=t}^m S_{\varrho,k} \right] \quad (99)$$

(a) substituting the discount values $\gamma_k = [[k \leq m]]$.

(b) because

$$\begin{aligned} & \sum_{h_{t:m} \in \mathcal{H}^m} P_\rho^\pi(h_{t:m} | h_{<t}) [S_{\varrho,k}] \\ = & \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\rho^\pi(h_{t:k} | h_{<t}) \sum_{h_{k+1:m} \in \mathcal{H}^{m-k+1}} P_\rho^\pi(h_{k+1:m} | h_{<t}h_{t:k}) [S_{\varrho,k}] \\ = & \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\rho^\pi(h_{t:k} | h_{<t}) [S_{\varrho,k}] \underbrace{\sum_{h_{k+1:m} \in \mathcal{H}^{m-k+1}} P_\rho^\pi(h_{k+1:m} | h_{<t}h_{t:k})}_{\leq 1} \end{aligned}$$

Where in the second line we used the chain rule of probability and in the last line we moved one summation on the right, since the surprise at step k , e.g. $-\log \varrho(o_k | h_{<k}a_k)$, does not depend on the future interactions sequence $h_{k+1:m}$

(c) because $P_\rho^\pi(h_{t:m} | h_{<t})$ does not depend on k .

Now we can distinguish the two cases, which are equivalent when the policy is deterministic:

Given $S_{\varrho,k} = -\log \varrho(o_k | h_{<t}a_t)$ we can write the following.

- **Entropy on observations**

From Eq. 97 we obtain Eq. 17:

$$\begin{aligned} S_{\rho||\varrho}^{\pi,m}(h_{<t}) &= \sum_{k=t}^m 1 \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\rho^\pi(h_{t:k} | h_{<t}) [-\log \varrho(o_k | h_{<k}a_k)] \\ &= \sum_{k=t}^m H_{\rho||\varrho}^\pi[\mathcal{O}_k | h_{<t}, \mathcal{H}_{t:k-1}, \mathcal{A}_k] \end{aligned} \quad (100)$$

- **Entropy of the chronological measure**

From Eq. 99 we obtain Eq. 18:

$$\begin{aligned} S_{\rho||\varrho}^{\pi,m}(h_{<t}) &\stackrel{(a)}{=} \sum_{h_{t:m} \in \mathcal{H}^m} P_\rho^\pi(h_{t:m} | h_{<t}) \left[-\log \prod_{k=t}^m \varrho(o_k | h_{<k}a_k) \right] \\ &\stackrel{(b)}{=} \sum_{h_{t:m} \in \mathcal{H}^m} P_\rho^\pi(h_{t:m} | h_{<t}) [-\log \varrho(o_{t:m} || a_{t:m}, h_{<t})] \end{aligned} \quad (101)$$

(a) because of the product rule of the logarithm and (b) by definition of chronological distribution.

Instead, with $S_{\varrho,k} = -\log P_\varrho^\pi(h_k | h_{<t}a_t)...$

- **Entropy over histories**

from Eq. 99 we get Eq. 19:

$$\begin{aligned} S_{\rho||\varrho}^{\pi,m}(h_{<t}) &= \sum_{h_{t:m} \in \mathcal{H}^m} P_\rho^\pi(h_{t:m} | h_{<t}) \left[-\log \prod_{k=t}^m P_\varrho^\pi(h_k | h_{<k}) \right] \\ &= \sum_{h_{t:m} \in \mathcal{H}^m} P_\rho^\pi(h_{t:m} | h_{<t}) [-\log P_\varrho^\pi(h_{t:m} | h_{<t})] \\ &= H_{\rho||\varrho}^\pi[\mathcal{H}^{m-t+1} | h_{<t}] \end{aligned} \quad (102)$$

Finally, we can combine entropy on histories (Eq. 102) and entropy on observations (Eq. 100).

- **H(histories) = H(observations)+H(actions)** First, from Eq. 100 and Eq. 101 we have:

$$\sum_{k=t}^m H_{\rho \parallel \varrho}^{\pi} [\mathcal{O}_k \mid h_{<t}, \mathcal{H}_{t:k-1}, \mathcal{A}_k] = \sum_{h_{t:m} \in \mathcal{H}^m} P_{\rho}^{\pi}(h_{t:m} \mid h_{<t}) [-\log \varrho(o_{t:m} \parallel a_{t:m}, h_{<t})]$$

Analogously, we can write:

$$\sum_{k=t}^m H_{\rho \parallel \varrho}^{\pi} [\mathcal{A}_k \mid h_{<t}, \mathcal{H}_{t:k-1}] = \sum_{h_{t:m} \in \mathcal{H}^m} P_{\rho}^{\pi}(h_{t:m} \mid h_{<t}) [-\log \pi(a_{t:m} \parallel o_{t:m-1}, h_{<t})]$$

Then, using again the product rule of log:

$$-\log \pi(a_{t:m} \parallel o_{t:m-1}, h_{<t}) - \log \varrho(o_{t:m} \parallel a_{t:m}, h_{<t}) = -\log P_{\varrho}^{\pi}(h_{t:m} \mid h_{<t})$$

Which gives us the desired equality:

$$H_{\rho \parallel \varrho}^{\pi} [\mathcal{H}^{m-t+1} \mid h_{<t}] = \sum_{k=t}^m H_{\rho \parallel \varrho}^{\pi} [\mathcal{O}_k \mid h_{<t}, \mathcal{H}_{t:k-1}, \mathcal{A}_k] + \sum_{k=t}^m H_{\rho \parallel \varrho}^{\pi} [\mathcal{A}_k \mid h_{<t}, \mathcal{H}_{t:k-1}]$$

from which we can easily obtain Eq. 16.

□

D Information Gain Plus Ambiguity Decomposition

In this section, we demonstrate how expected surprise can be decomposed as information gain plus ambiguity (Lem. 6.1. We'll use often the definition of posterior:

$$w_\nu(h_{<t}) = \frac{w_\nu \nu(o_{<t} \| a_{<t})}{\xi(o_{<t} \| a_{<t})} = \frac{w_\nu P_\nu^\pi(h_{<t})}{P_\xi^\pi(h_{<t})} \quad (103)$$

We can now write the full expression for the expected surprise of the **sma**($\xi \| \xi$) agent.

$$\begin{aligned} S_{\xi \| \xi}^{\pi, \gamma}(h_{<t}) &= \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\xi^\pi(h_{t:k} | h_{<t}) S_{\xi, t} \\ &= \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\xi^\pi(h_{1:k}) S_{\xi, t} \\ &= \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \sum_{\nu \in \mathcal{M}} w_\nu P_\nu^\pi(h_{1:k}) S_{\xi, t} \end{aligned}$$

Now, expanding the point-wise cost with the two alternative version as negative log conditional likelihood on observation (Eq. S[o]) or action-observation pair (Eq. S[h]):

$$\begin{aligned} &= \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \sum_{\nu \in \mathcal{M}} w_\nu P_\nu^\pi(h_{1:k}) \log \left[\frac{\xi(o_{1:k-1} \| a_{1:k-1})}{\xi(o_{1:k} \| a_{1:k})} OR \frac{P_\xi^\pi(h_{1:k-1})}{P_\xi^\pi(h_{1:k})} \right] \\ &\stackrel{(a)}{=} \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \sum_{\nu \in \mathcal{M}} w_\nu P_\nu^\pi(h_{1:k}) \left[\log \frac{w_\nu(h_{1:k})}{w_\nu(h_{1:k-1})} + \log \left[\frac{\nu(o_{1:k-1} \| a_{1:k-1})}{\nu(o_{1:k} \| a_{1:k})} OR \frac{P_\nu^\pi(h_{1:k-1})}{P_\nu^\pi(h_{1:k})} \right] \right] \\ &\stackrel{(b)}{=} \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \sum_{\nu \in \mathcal{M}} w_\nu P_\nu^\pi(h_{1:k}) \left[\log \frac{w_\nu(h_{1:k})}{w_\nu(h_{1:k-1})} + S_{\nu, k} \right] \\ &\stackrel{(c)}{=} \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \sum_{\nu \in \mathcal{M}} w_\nu P_\nu^\pi(h_{1:k}) \left[\log \frac{w_\nu(h_{1:k})}{w_\nu(h_{1:k-1})} \right] + \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \sum_{\nu \in \mathcal{M}} w_\nu P_\nu^\pi(h_{1:k}) [S_{\nu, k}] \quad (104) \\ &\stackrel{(d)}{=} \frac{1}{P_\xi^\pi(h_{<t})} \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\xi^\pi(h_{1:k}) \sum_{\nu \in \mathcal{M}} w_\nu(h_{1:k}) \frac{w_\nu(h_{1:k})}{w_\nu(h_{1:k-1})} + \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \sum_{\nu \in \mathcal{M}} \frac{w_\nu}{P_\xi^\pi(h_{<t})} P_\nu^\pi(h_{1:k}) S_{\nu, k} \\ &\stackrel{(e)}{=} \mathbb{E}_\xi^\pi [IG^\gamma | h_{<t}] + \sum_{\nu \in \mathcal{M}} w_\nu(h_{<t}) \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} \frac{P_\nu^\pi(h_{1:k})}{P_\xi^\pi(h_{<t})} S_{\nu, k} \\ &= \mathbb{E}_\xi^\pi [IG^\gamma | h_{<t}] + \sum_{\nu \in \mathcal{M}} w_\nu(h_{<t}) S_{\nu \| \nu}^\pi(h_{<t}) \end{aligned}$$

In (a) the inverse of the posterior formula (Eq. 103) was used on both numerator and denominator, w_ν cancels out. In (b) we collapsed the alternative formulations of negative log likelihood, given predictor ν , back to the agnostic notation $S_{\nu, k}$. In (c) we split the summation in two identical summations for each of the two addends in the square brackets. In (d) we moved the fraction $\frac{1}{P_\xi^\pi(h_{<t})}$ to be the inner sum of the right term, while we used the equality $w_\nu P_\nu^\pi(h_{1:k}) = w_\nu(h_{1:k}) P_\xi^\pi(h_{1:k})$ (Eq. 103) whin the term on the left. The latter was then substituted in (e) with the definition of expected information gain (Eq.3), while, on the right, we used the equality $\frac{w_\nu}{P_\xi^\pi(h_{<t})} = \frac{w_\nu(h_{<t})}{P_\nu^\pi(h_{<t})}$ (Eq. 103) and moved the sum over posterior weight to be the outer sum. Finally, with the last equation we substituted in the the definition of expected surprise $S_{\nu \| \nu}^{\pi, \gamma}$.

E IG and ν -specific IG as KL divergence

It was already shown in [Ors13] that the expected information gain is equal to weighted cumulative KL divergence between each model $\nu \in \mathcal{M}$ and the mixture ξ . Along with the latter, here we show the equality $IG_\nu = KL_{\nu||\xi}$. Continuing from the formulation in 104 and using the equality $\frac{w_\nu}{P_\xi^\pi(h_{<t})} = \frac{w_\nu(h_{<t})}{P_\nu^\pi(h_{<t})}$ (Eq. 103):

$$\mathbb{E}_\xi^\pi [IG^\gamma | h_{<t}] = \sum_{\nu \in \mathcal{M}} w_\nu(h_{<t}) \underbrace{\sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\nu(h_{t:k} | h_{<t}) \left[\log \frac{w_\nu(h_{1:k})}{w_\nu(h_{1:k-1})} \right]}_{\mathbb{E}_\nu^\pi [IG_\nu^\gamma | h_{<t}]} \quad (105)$$

Where the highlighted portion represents a model-specific IG, which quantifies the expected cumulative increase of belief in ν , encoded by $\log \frac{w_\nu(h_{1:k})}{w_\nu(h_{1:k-1})}$, if observations were effectively generated by ν with $P_\nu(h_{t:k} | h_{<t})$ (the belief can only increase). Continuing the above:

$$\dots = \sum_{\nu \in \mathcal{M}} w_\nu(h_{<t}) \mathbb{E}_\nu^\pi [IG_\nu^\gamma | h_{<t}]$$

Now we show $\mathbb{E}_\nu^\pi [IG_\nu^\gamma | h_{<t}] = KL_h^\gamma(P_\nu^\pi || P_\xi^\pi)$. The derivation is straightforward using the posterior formula in the log term, from Eq. 105:

$$\begin{aligned} \mathbb{E}_\nu^\pi [IG_\nu^\gamma | h_{<t}] &= \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\nu(h_{t:k} | h_{<t}) \left[\log \frac{w_\nu(h_{1:k})}{w_\nu(h_{1:k-1})} \right] \\ &= \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\nu(h_{t:k} | h_{<t}) \left[\log \frac{w_\nu P_\nu^\pi(h_{1:k}) P_\xi^\pi(h_{1:k-1})}{w_\nu P_\nu^\pi(h_{1:k-1}) P_\xi^\pi(h_{1:k})} \right] \\ &= \sum_{k=t}^{\infty} \gamma_k \sum_{h_{t:k} \in \mathcal{H}^{k-t+1}} P_\nu(h_{t:k} | h_{<t}) \left[\log \frac{P_\nu^\pi(h_k | h_{<k})}{P_\xi^\pi(h_k | h_{<k})} \right] = KL_{h_{<t}}^\gamma(P_\nu^\pi || P_\xi^\pi) \end{aligned} \quad (106)$$

Which combined with Eq. 105 also proves the already known:

$$\mathbb{E}_\xi^\pi [IG^\gamma | h_{<t}] = \sum_{\nu \in \mathcal{M}} w_\nu(h_{<t}) KL_{h_{<t}}^\gamma(P_\nu^\pi || P_\xi^\pi)$$

E.1 Telescoping IG and KL

Given $h_{<t} = \epsilon$ and $\gamma_k = \mathbb{I}[k \leq m]$ we can telescope ν -specific IG and, as already know, KL divergence too with a similar reasoning as in Sec. C. Considering an empty history $h_{<t} = \epsilon$ for simplicity, we can write:

$$\mathbb{E}_\nu^\pi [IG_\nu^m | \epsilon] = \sum_{h_{1:m} \in \mathcal{H}^m} P_\nu^\pi(h_{1:m}) \log \frac{w_\nu(h_{1:m})}{w_\nu} = \sum_{h_{1:m} \in \mathcal{H}^m} P_\nu^\pi(h_{1:m}) \log \frac{P_\nu^\pi(h_{1:m})}{P_\xi^\pi(h_{1:m})} = KL_\epsilon^m(P_\nu^\pi || P_\xi^\pi)$$

Since $\log \frac{1}{w_\nu} = K(\nu)$ we can also re-write the expected ν -specific IG as follows:

$$\mathbb{E}_\nu^\pi [IG_\nu^m | \epsilon] = K(\nu) - \mathbb{E}_\nu^\pi [-\log w_\nu(h_{1:m})] \stackrel{(a)}{\leq} K(\nu) \quad (107)$$

(a) because the value $w_\nu(\cdot)$ can only increase up to one and $-\log 1 = 0$, this means that when the KL divergence is maximal, the posterior will have converged fully on ν . We could quantify ν -specific IG further in the deterministic setting, considering it has value $w_\nu(h) = \frac{w_\nu}{\mathbf{W}_\nu(h)}$. On the other hand, we could also directly quantify the KL divergence:

$$KL_\epsilon^m(P_\nu^\pi || P_\xi^\pi) = S_{\mu||\xi}^{\pi, m} = -\log \xi(o||a) \stackrel{(a)}{\pm} K(o | a) \quad o = \nu(a), a \sim \pi \quad (108)$$

Where (a) follows from Thm. 7.6. Finally, with the above we could then leverage the lower bound we proved in 8.1.

F Coding Theorem for Deterministic Environments

The following is the proof for Thm. 7.6.

Proof. Using Eq. 18 we have that:

$$\begin{aligned} S_{\mu||\xi}^{\pi,m}(\epsilon) &= \sum_{h' \in \mathcal{H}^m} P_{\mu}^{\pi}(h') [-\log \xi(o' || a')] \\ &= -\log \xi(o || a) \end{aligned} \quad (109)$$

Where o, a in $h_{1:m} \sim P_{\mu}^{\pi}(\cdot)$ is the interaction history generated deterministically by μ (which $\mathbf{sma}(\mu || \xi)$ knows) and π .

Then, since for each a we can encode the observation trajectory o with a prefix code $C_a(o) = c$ given the probability distribution $\xi(o || a)$ with upper semi-computable length $l(c) \leq -\log \xi(o || a) + 3$ using the construction detailed in [Li19, Lem. 4.3.3, Pg.276, 277]. This construction considers a slightly different distribution, $\xi(x | y)$, where x is the possible continuation of the output stream y ²⁹, but the proof doesn't make any assumptions on the distribution, it only accounts for lower semi-computability which holds for $\xi(o || a)$ too.

Since $K(o | a)$ gives the shortest prefix code for o given a we have that [Li19, Thm. 4.3.4]:

$$K(o | a) \leq -\log \xi(o || a) + 3 \quad (110)$$

Then, considering that monotone complexity, Km , is lower than prefix-complexity:

$$K(o | a) \geq Km(o | a) \stackrel{\pm}{=} Km_{U'}(o | a) \quad (111)$$

Where U' is a monotone UTM and equality holds up to an additive constant holds by the invariance Theorem [Hut24, Thm. 2.7.6]. Then we have that trivially:

$$2^{-Km_{U'}(o|a)} \leq M(o || a) = \sum_{p: U'(p,a) \rightarrow o*} 2^{-l(p)} \quad (112)$$

↓

$$Km_{U'}(o | a) \geq -\log M(o || a) \quad (113)$$

Combining Eq. 111 and 113 we obtain:

$$K(o | a) \stackrel{\pm}{\geq} -\log M(o || a) \quad (114)$$

Now we demonstrate that $M(o || a) \geq c \cdot \xi(o || a)$. This stems from the fact that we can represent any minimal program p running on U' as a deterministic environment $\nu_{p'} = U'(p) \in \mathcal{M}$, with $K(\nu_{p'}) \leq l(p) + O(1)$. In fact, by definition, the set $\{p : U'(p, a) \rightarrow o*\}$ contains the programs for which the monotone machine U' stops reading the input tape containing the program. Hence, it won't read longer programs $\{q \in \mathbb{B}^* : p \sqsubset q\}$ [Hut24, Sec. 3.8.1]. This means that the programs considered in the summation (Eq. 112) form a prefix free set. This allows us, given any action sequence, to find a matching model $\nu_{p'} \in \mathcal{M}_{\mu}^{\pi,m}(\epsilon)$ with $p' = u'p$, $U'(p, a) = U(u'p, a)$, $K(\nu_{p'}) = K(p')$. Then we can construct T_i as an identity machine for the prefix set of minimal programs with

²⁹On the other hand, given the chronological distribution $\xi(o || a)$ (or $M(o || a)$) we have to consider actions given by the policy distribution (on a separate additional input tape) since the environment distributions $\nu(\text{programs})$ produce only observations and actions are externally defined by the AIXI algorithm, which here can be simplified to a deterministic sequence of actions, but generally we would have to consider a policy π

input a $U(i'a'p) = T_i(a'p) = p$ for each program, where, when run on U' with actions a . This gives us:

$$\begin{aligned} K(\nu_p) &\leq l(i') + l(a') + l(p) \\ &= l(p) + O(1) \end{aligned} \quad (115)$$

Since $l(a') = l(a) + \log l(a)$ is constant (for a fixed horizon m).



We might be able to get rid of the actions overhead, or reduce it a simple log term, in other ways too. Intuitively, it allows to find the right mapping between programs p and models ν_p , but does the length of the action sequence increase the complexity of the programs that have interact with such sequences? If that's not true we don't need the overhead, if it is: it would probably increase the length $l(p)$ of the program p running on U' , which already bounds $K(\nu_p)$.

Finally, we can write:

$$\begin{aligned} \xi(o || a) &= \sum_{\nu_{p'} \in \mathcal{M}_{\mu}^{\pi,m}} 2^{-K(\nu_p)} \\ &\stackrel{(a)}{\geq} \sum_{\nu_{p'} \in \mathcal{M}_{\mu}^{\pi,m}} 2^{-c} 2^{-l(p)} \\ &= 2^{-c} \sum_{\nu_{p'} \in \mathcal{M}_{\mu}^{\pi,m}} 2^{-l(p)} \\ &\stackrel{(b)}{\geq} c_0 \sum_{p: U'(p,a) \rightarrow o*} 2^{-l(p)} = c_0 M(o || a) \end{aligned}$$

The inequality (a) follows from Eq. 115, while with (b) we renamed the multiplicative constant $c_0 = 2^{-c}$ and we substituted the summation set. In fact, there is a mapping $\{p : U'(p, a) \rightarrow o*\} \rightarrow \{\nu \in \mathcal{M}_{\mu}^{\pi,m}\}$ as shown above.

With this we obtained

$$\xi(o || a) \geq c_0 M(o || a). \quad (116)$$

Applying the negative logarithm on both sides

$$-\log \xi(o || a) \leq -\log M(o || a) + c \quad (117)$$

With the latter, Eqs. 110 and 114, we can collapse the cycle of inequalities into equality up to additive constants and obtain:

$$S_{\mu||\xi}^{\pi,m} \stackrel{(a)}{=} \xi(o || a) \stackrel{\pm}{=} M(o || a) \stackrel{\pm}{=} K(o | a) \quad (118)$$

with

$$o, a \text{ in: } h \sim P_{\mu}^{\pi}(\cdot)$$

(a) by definition as shown in Eq. 109. □

The last two equalities are a standard result and usually demonstrated in a similar fashion [Li19, Thm. 4.3.3]. Nonetheless, as already hinted, they are proven in the context of models/machines which generate observations, without the AIXI interaction cycle. And the considered distribution $\xi(o)$ or $\xi(o' | o)$ over trajectories can be defined without any conditioning on the actions. On the other hand, here they necessarily interleave each observation and have to be accounted for when

computing the probability of a future trajectory: through the chronological distribution $\xi(o||a)$.³⁰ Nevertheless, if we considered the model class $\mathcal{M}_{sol}$ of semi-measures, the proof would have been pretty similar and straightforward, as a similar result has been mentioned in [Hut05, Eq. 1.10] and an even stronger one has been proven by [Woo13], where the equivalence between the class of Solomonoff priors and universal mixtures was demonstrated. Here, instead, we are considering a model class of deterministic environments, this doesn't change considerably the first two thirds of the proof, except for the last part which shows the dominance $\xi \geq cM$ (Eq. 116). The latter is not obvious, since ξ was made "less powerful" by limiting its model class to $\mathcal{M}_{det}$ (Sec. 2.4). In fact, proving this usually relies on the fact that $M(\cdot) \in \mathcal{M}_{sol}$.

G Proportion of c-Incompressible Environments

The following is a (failed) proof attempt which suggests that the opposite might be true instead: most environments are c-compressible, meaning they admit a low complexity niche $\min_a K(\mu(a) | a) \stackrel{\pm}{=} K(\mu') \leq l(o) - c$.

Conjecture G.1 (Most environments are c-incompressible). There are at least $2^{l(o)}(1 - 2^{l(a)+c_\mu-c})$ environments in the class $\mathcal{M}_{\leq l(o)}$ ($K(\mu) \leq l(o)$) such that $K(\mu') \geq l(o) - c$. This means that the proportions of c-incompressible environments is $(1 - 2^{l(a)+c_\mu-c})$ since $|\mathcal{M}_{\leq l(o)}| \leq 2^{l(o)}$.

Proof. This might be obtained considering environments constructed as in Lem. 8.2 and using the incompressibility of strings theorem [Li19, Thm. 2.2.1] which states at least $2^{l(o)}(1 - 2^{-c})$ strings of length $l(o)$ have $K(o) \geq l(o) - c$. For each of such strings, let's call it o_{μ^*} , we have a unique environment μ constructed as in Lemma 8.2 with $K(\mu^*) \stackrel{\pm}{=} K(o^*)$.

We rewrite the bound from Thm. 8.1 and require an additional lower bound $l(o) - c$ and fix c_μ as our "action-dependency budget" (which does not depend on μ^*).

$$\begin{aligned} l(o) - c &\leq K(\mu^*) - l(a) - 2\log l(a) - c_\mu \\ &\leq K(\mu) - l(a) - 2\log l(a) - c_\mu \\ &\leq K(\mu') \end{aligned}$$

This means

$$\begin{aligned} l(o) - c + l(a) + 2\log l(a) + c_\mu &\leq K(\mu^*) \\ &\leq K(\mu) \leq l(o) \end{aligned}$$

This means that to count the possible μ^* which provide a lower bound for $K(\mu')$ greater than $l(o) - c$ we can count the number of programs with complexity between $l(o) - c' = l(o) - c + l(a) + 2\log l(a) + c_\mu$ and $l(o)$.

³⁰Alternatively, we could have considered the distribution over histories $P_\xi^\pi(h)$. This might have complicated things as we would have had to change the definition of M to generate actions too. For example, by adding a program q to generate actions, similarly to what was done for earlier definitions of AIXI [Hut07]. Moreover, this would have led to results in term of complexity of observations and actions entangled in histories: $K(h)$ or $K(a, o)$ in the best case. The latter might also be desirable, but I thought it was cleaner to talk in terms of $K(o | a)$, which relates to Thm. 8.1 and fits well with the intuition that SMA wants to preserve the simplest model or niche which generates o given a . Additionally, $\xi(o||a) = P_\xi^\pi(h)$ when the policy is deterministic, as already mentioned.

Since $K(o_{\mu^*}) \stackrel{\pm}{=} K(\mu^*)$ and there is an bijective relationship $o_{\mu^*} \leftrightarrow \mu^*$, we can also count the number of strings of length $l(o)$ with complexity $l(o) - c \leq K(o_{\mu^*}) \leq l(o)$

In order to do this we can proceed similarly to the proof in [Li19, Thm. 2.2.1]: we count the programs with length $\leq l(o) - c'$ and then subtract it from the total number of strings $\leq l(o)$. The number of programs with length strictly less than $l(o) - c'$ is

$$\sum_{i=0}^{l(o)-c'-1} 2^i = 2^{l(o)-c'} - 1 \quad (119)$$

Hence, there are at least $2^{l(o)} - 2^{l(o)-c'} o_\mu^*$ each with no program of length less than $l(o) - c'$ and a unique μ^* such that $l(\mu^*) = K(o_\mu^*)$. $\square$

 The problem here is that we would have strings with length $l(o)$ but with higher complexity $l(o) + 2\log l(o) \geq K(\mu^*) \stackrel{\pm}{=} K(o_{\mu^*}) \geq l(o)$. Furthermore, we cannot have the complexity of be μ^* close or equal to $l(o)$, again, this would entail $K(\mu) \geq l(o)$. We could increase the size of the environments class considered, but this in turn would shrink the proportion of c-incompressible environments, it is a dog biting its own tail. It might be easier to proving the opposite in the general case. Indeed, we might be able to prove a similar statement within the class of environments built with a specific construction, such as the one used in Lem. 8.2. In the class, while i^+ is fixed, j can be any sequence of $l(j)$ bits as long as it produces an observation as we already know it will be compressed in the worst case and we are counting for it as we are subtracting the compressibility bound by c_μ . Finally, a big proportion of minimal programs μ^* give the required complexity. This would still be quite interesting considering Prop. 8.3.

H Tightness of Lower Bound

Conjecture 8.1. The lower bound presented in Thm. 8.1, here named "LB", is tight.

$$K(\mu') - LB \stackrel{\pm}{\leq} l(a) + 2\log l(a)$$

Where:

$$LB = K(\mu) - l(a) - 2\log l(a) - c_\mu \leq K(\mu | a) - c_\mu \stackrel{\pm}{=} K(\mu') \quad (59)$$

Proof. In fact, the bound does not depend on the action sequence, while $K(o | a)$ does and, as we'll see, it is minimized by $\mathbf{sma}(\mu||\xi)$ (Cor. 7.7). For instance, in order to minimize the complexity of observations, the agent could pick the actions which lead to a low (action dependent) $K(o_j) = O(1)$, while it cannot directly affect the incompressible action-independent component o_{μ^*} . Then, we can write:

$$\begin{aligned} K(\mu') &\stackrel{\pm}{=} \min_a K(o | a) \stackrel{\pm}{\leq} K(\mu^* | a) + \underbrace{l(i^+)}_{=O(1)} + \underbrace{K(o_j)}_{=O(1)} \\ &\stackrel{\pm}{=} K(\mu | a) - K(j, i^+ | \mu^*, K(\mu^*)) \end{aligned} \quad (120)$$

$$\stackrel{(a)}{\approx} K(\mu | a) - c_\mu \quad (121)$$

Finally, applying the equality in Eq. 52 along with As. III (Eq. 58) and Lem. 8.2. we obtain

$$K(\mu | a) - c_\mu \leq \min_a K(o | a) \stackrel{\pm}{=} K(\mu')$$

The approximation (a) is done assuming $K(j, i^+ | \mu^*, K(\mu^*)) \approx l(i^+ j) \stackrel{*}{=} c_\mu$: μ^* contains little information useful to generate j (and i^+). This could be motivated by the fact that μ^* is the minimal action independent program to generate a specific sequence o_μ^* , while j could be any action dependent program.

☀ Even without this clean bound we already have gauranteed tightness because of the $K(\mu)$ upper bound for any policy :

$$K(\mu) - l(a) - 2 \log l(a) - c_\mu \leq K(\mu') \leq K(\mu) \quad (122)$$

□

I The Agnostic Agent Explores Accidentally

Example I.1. Let's say the true environment is quite complex: $K(\mu) \gg 0$, and constructed as in Lem. 8.2. Then we have a bounded number of possible niches:

$$\left| \bigcup_{\pi} \{ \mu' \in \mathcal{M}_{\leq \mu}^{m, \pi} : K(\mu | a) - c_\mu \leq K(\mu') \leq K(\mu | a) \} \right| \stackrel{(a)}{\leq} \quad (123)$$

$$|\mathcal{A}^m| 2^{c_\mu} \stackrel{(b)}{\leq} 2^{l(a) + 2l(i^+ j) + c_1} \quad (124)$$

(a) Since μ can produce up to $|\mathcal{A}^m|$ distinct observation sequences, each with a given minimal program $\mu_a' : K(o | a) = l(\mu') \stackrel{\pm}{=} K(\mu'_a)$ ³¹ Then, since $K(\mu | a) - c_\mu \leq K(o | a) \stackrel{\pm}{=} K(\mu'_a) \leq K(\mu') \leq K(\mu | a)$, we can use the fact that the number of programs p which produce the same observation sequence x with $K(x) \leq K(p) \leq K(x) + c$ are bounded by 2^c :

$$\forall a \sim \pi \quad (125)$$

$$\left| \{ \mu' \in \mathcal{M}_{\leq \mu}^{m, \pi} : K(\mu | a) - c_\mu \leq K(\mu') \leq K(\mu | a) \} \right| \leq 2^{c_\mu} \quad (126)$$

(b) holds by $c_\mu \leq 2l(i^+ j) + c_1$ (Eq. 69).

On the other hand, the bound on the smaller, infeasible, niches is much bigger:

$$\begin{aligned} |\mathcal{M}_{\leq K(\mu')}| &\stackrel{(c)}{\lesssim} 2^{K(\mu) - l(a) - 2 \log l(a) - c_\mu} \\ &\stackrel{(d)}{\leq} 2^{l(o) + 2 \log l(o)} \end{aligned}$$

(c) assuming the bound on $K(\mu')$ is tight (Conj. 8.1). Otherwise, the number on unfeasible niches could be even higher. (d) was added as a sanity check and holds because $K(\mu') \leq K(o) \leq l(o) + 2 \log l(o)$. Furthermore, **sma**($\xi || \xi$) doesn't know

³¹ μ' is what ultimately determines the value $S_{\mu || \xi}^\pi$.

$$\begin{aligned} \forall a \sim \pi, o = \mu(a) \exists \mu'_a = \nu_{p_a^*} : \\ K(\mu'_a) \stackrel{\pm}{=} \underbrace{K(o | a)}_{\stackrel{\pm}{=} S_{\mu || \xi}^\pi} \stackrel{\pm}{=} K(\mu') \quad \forall \mu' \in \mathcal{M}_\mu^\pi \end{aligned}$$

Where (a) is motivated by Eq. 46.

all these are unfeasible before it has sampled the $l(o)$ bits of the observation sequence from μ (Eq. 77).

Therefore, at each timestep the agent will optimize considering with exponentially higher weight these simpler environments.

We can then hypothesize that, from the perspective of μ , the action sequence will look almost random. Now, let's assume that, on average, for interaction cycle there is, on average at most one action that allows a potential niche to output the same observation as μ for it to be kept within the models pool (survival probability of $\varepsilon \leq \frac{1}{|\mathcal{A}|}$). This is a fairly conservative estimate if we consider $|\mathcal{A}| \ll |\mathcal{O}|$. This means that each potential niche has a survival probability over the full horizon of

$$|\mathcal{O}|^{-m} \leq \varepsilon^m \leq |\mathcal{A}|^{-m} = 2^{\log[|\mathcal{A}|^{-m}]} = 2^{-l(a)}$$

. Where ε is an arbitrary per-step probability of survival above $|\mathcal{O}|^{-1}$ (survival if prob. of observation sequences were uniform over programs). Under this assumption, the bound on the expected number of non-discarded niches is obtained by multiplying the survival probability with the bound in 124:

$$\mathbb{E}_\mu^{\pi_\xi || \xi} |\mathcal{M}_{\leq \mu}^{\pi_\xi || \xi}| \leq 2^{c_1 + 2l(i^+ j)} \quad (127)$$

This value seems quite high, on the other hand if $\varepsilon = |\mathcal{A}|^{-2}$ so that $\varepsilon^{-m} = 2^{2l(a)}$ and $l(a) \geq 2l(i^+ j) + c_1$ the expected number of surviving niches would shrink to 1. Furthermore, simpler ones have a lower chance of being kept since they come in lower number. With this idea and given :

$$2^{l(a)} 2^{m \log \varepsilon} = 2^{m [\log |\mathcal{A}| - \log \frac{1}{\varepsilon}]} = 2^{-m \log \frac{1}{\varepsilon_1}} = 2^{-m E_1} \quad (128)$$

with $E_1 = \log[|\mathcal{A}| \varepsilon]$ we can update the bound (Eq. I.1) of the number of non-discarded niches at least c_2 bits simpler than μ as:

$$\mathbb{E}_\mu^{\pi_\xi || \xi} |\mathcal{M}_{\leq \mu - c_2}^{\pi_\xi || \xi}| \leq 2^{c_1 + 2l(i^+ j) - c_2 - m E_1}$$

The statement we can make is even stronger if $K(\mu | a) \approx K(\mu)$ suggesting that **SMA** will drop all niches with complexity $\leq K(\mu | a) - c_2 \approx K(\mu) - c_2$ with probability $1 - 2^{-c_2}$ if $m E_1 = c_1 + 2l(i^+ j)$.

In the extreme case where all niches with $K(\mu') \leq K(\mu)$ are dropped, the Solomonoff predictor will weight the true environment μ the most as it can't be discarded by definition, out of all remaining models in the pool.

★ It is important to note that in the example presented above several assumption were made and the conclusion does not hold in general. Mainly, we considered a random action sequence and assigned an arbitrary probability of survival for every niche (on average). Nonetheless, there are several ($2^{l(a)}$) niches, namely μ'_a (Lem. 7.4), that have each a probability of survival of at least $2^{-l(a)}$, as they will be surely kept if the corresponding action sequence, for which they are the minimal niche, is selected. How the **sma**($\xi || \xi$) agent behaves in general is still an open problem.

I.1 Additional Ideas

Conjecture I.1. $\mathbf{sma}(\xi||\xi)$ is asymptotically optimal in mean if the complexity of the true environment is finite:

$$\begin{aligned} &\text{if } \exists N \in \mathbb{N}^+ : K(\mu) \leq N \implies \\ &\lim_{t \rightarrow \infty} \mathbb{E}_{\mu}^{\pi_{\xi||\xi}} \left[S_{\mu||\xi}^{\pi_{\xi||\xi}}(h_{<t}) - S_{\mu||\xi}^{\pi_{\mu||\xi}}(h_{<t}) \right] = 0 \end{aligned}$$

Proof. This could be proven similarly to how optimality was proven for deterministic KSA[Ors14].

To describe our intuition, we first consider the case where SMA finds a minimal niche with complexity $K(\mu') \leq K(\mu)$ in which it could stay indefinitely. Inside this niche no considerable new information is gathered and the minimal niche dominates the cost (Cor. 7.7). Therefore, $\mathbf{sma}(\xi||\xi)$ has no interest in leaving the niche³², $\mathbf{sma}(\mu||\xi)$ wouldn't either as discarding μ' is surely a suboptimal action knowing μ . Instead, if no stable niche is found, μ itself will become the simplest environment surviving in the posterior and we can apply the same reasoning as above. The constrained computability of μ serves to counter the fact that after an arbitrary number of time steps the true environment could, quite literally, surprise the agnostic agent by producing observations different from expected infinitely often. It can do it several times, and $\text{AI}\xi$ would jump from the previous to the new minimal niche, but it cannot happen infinitely often, since ξ will eventually discard all $2^N - 1$ possible programs. If the agent discarded more programs, it would mean that no environment with complexity below the threshold can describe the observation, and μ would have to be more complex, contradicting the hypothesis. In summary, μ cannot surprise the $\text{AI}\xi$ agent indefinitely, after some time step $t \geq N_1$ this can't happen anymore if the complexity of μ is limited. $\square$

³²this has to be demonstrated and we can't take it for granted.